

Promotions & Bundles Management

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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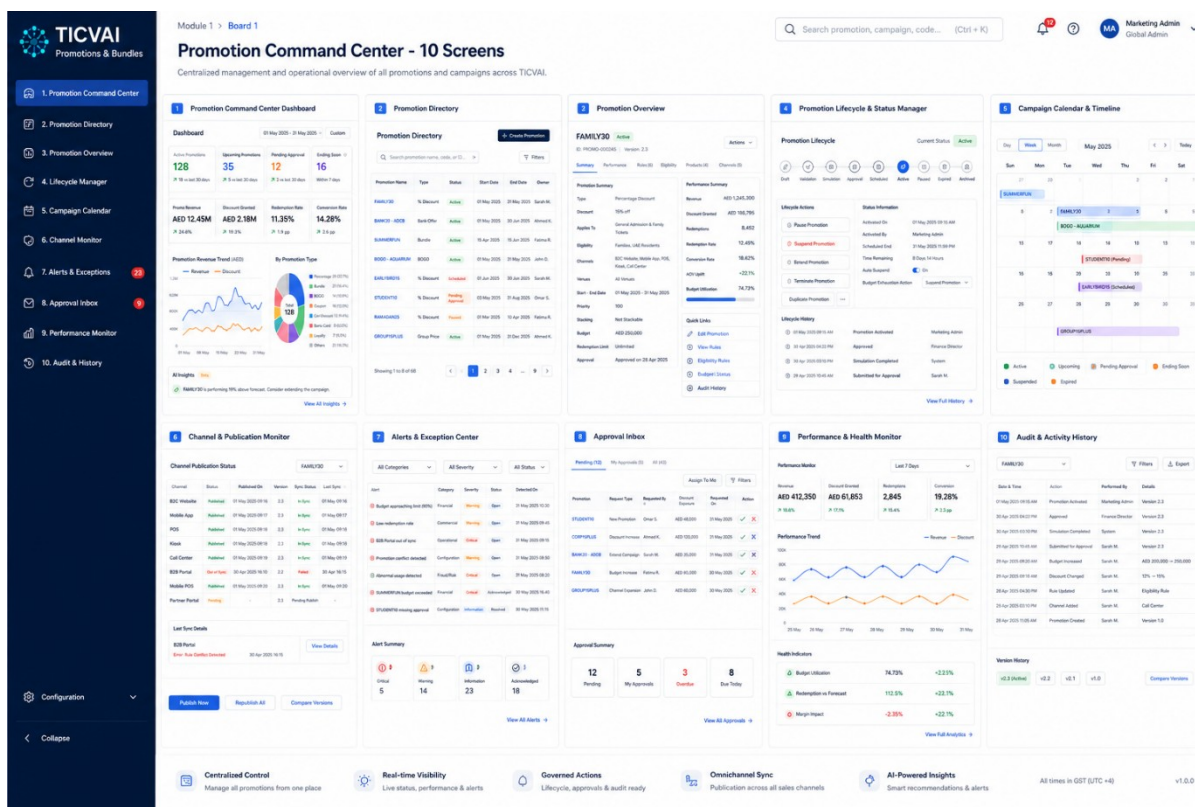
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Module 1 — Promotions & Bundles Management

Board 1 — Promotion Command Center



Board Objective

Provide a centralized back-office command center for managing the complete lifecycle and operational status of promotions, discounts, coupons, bundles, BOGO offers, partner campaigns, payment-method offers, and AI-generated promotional recommendations.

The board shall allow authorized business users to:

- See all promotional activities across TICVAI.
- Search and filter promotions.

- Identify active, upcoming, expired, suspended, draft, and approval-pending campaigns.
- Understand campaign financial exposure and performance.
- Detect issues, conflicts, budget overruns, and approaching limits.
- Start promotion creation.
- Duplicate or version existing promotions.
- Submit, approve, reject, activate, pause, or terminate promotions.
- Monitor omnichannel publication.
- Access campaign audit history.
- Receive AI recommendations and alerts.

This board becomes the **entry point into the complete Promotions & Bundles module.**

Screen 1 — Promotion Command Center Dashboard

Purpose

Provide management with a real-time executive and operational overview of all promotional activities.

Main Components

Global KPI cards

- Active Promotions
- Upcoming Promotions
- Draft Promotions
- Pending Approval
- Suspended Promotions
- Promotions Ending Soon
- Total Promotion Revenue
- Total Discount Granted
- Incremental Revenue
- Promotion Conversion Rate
- Redemption Rate
- Average Order Value Uplift
- Promotion Cost
- Estimated Margin Impact
- Campaign Budget Utilization

Promotion Type Summary

Break down current promotions by:

- Percentage discount
- Fixed-value discount
- BOGO
- Volume discount
- Cart-level discount

- Coupon
- Promo code
- Bundle
- Combo
- Added-value promotion
- Free product
- Free ticket
- Payment-method promotion
- Bank/card promotion
- Membership promotion
- Loyalty promotion
- Partner promotion
- Group promotion
- Flash sale
- Early-bird
- Last-minute
- Dynamic/AI promotion

Dashboard Visualizations

- Promotion revenue trend
- Discount exposure trend
- Conversion uplift
- Promotions by channel
- Promotions by venue
- Promotions by product
- Campaign budget consumption
- Top-performing promotions
- Underperforming promotions

AI Panel

AI shall highlight items such as:

- Promotion performing above forecast.
 - Promotion creating excessive discount exposure.
 - Conversion improving without corresponding margin deterioration.
 - Campaign likely to exceed budget.
 - Promotion with low redemption.
 - Promotion potentially cannibalizing another campaign.
 - Recommended action: extend, pause, modify, increase, decrease, or test.
-

Screen 2 — Promotion & Campaign Directory

Purpose

Provide the master searchable list of every promotion and commercial campaign created within TICVAI.

Table Columns

- Promotion ID
- Promotion Name
- Promotion Type
- Campaign
- Status
- Business Entity
- Venue
- Product
- Target Segment
- Channel
- Start Date
- End Date
- Discount Type
- Discount Value
- Budget
- Redemption Count
- Revenue Generated
- Owner
- Approval Status
- Version
- Last Modified

Supported Statuses

- Draft
- Configuration Incomplete
- Simulation Required
- Pending Approval
- Approved
- Scheduled
- Active
- Paused
- Suspended
- Budget Exhausted
- Expired
- Cancelled
- Archived

Filters

Allow filtering by:

- Promotion type
- Campaign
- Product
- Product category
- Venue
- Attraction
- Event
- F&B
- Retail
- Membership
- Channel
- Partner
- Customer segment
- Date
- Status
- Owner
- Approval state
- Promotion value
- Budget status

Bulk Actions

Authorized users may:

- Activate
 - Pause
 - Extend
 - Duplicate
 - Archive
 - Export
 - Assign owner
 - Submit for approval
-

Screen 3 — Promotion Overview

Purpose

Provide the complete business summary of one selected promotion without opening its detailed configuration.

Header

Display:

- Promotion name
- Promotion ID

- Version
- Status
- Owner
- Created by
- Effective dates
- Campaign
- Business entity

Promotion Summary

Show:

- Promotion mechanism
- Discount/reward
- Eligible products
- Eligible customers
- Eligible channels
- Eligible locations
- Applicable time/date conditions
- Usage limits
- Budget
- Redemption ceiling
- Promotion hierarchy
- Stacking behavior
- Approval state

Performance Summary

Show:

- Transactions
- Redemptions
- Gross revenue
- Discount granted
- Net revenue
- Incremental revenue
- Conversion
- AOV impact
- Profit/margin impact
- Remaining budget

Navigation

Direct links to:

- Rule configuration
- Eligibility
- Products
- Channels

- Codes/coupons
 - Budget
 - Simulation
 - Approval
 - Analytics
 - Audit history
-

Screen 4 — Promotion Lifecycle & Status Manager

Purpose

Control the operational lifecycle of promotions.

Lifecycle

Draft → Validation → Simulation → Approval → Scheduled → Active → Paused/Suspended → Expired → Archived

Available Actions

Depending on user permissions:

- Save Draft
- Validate
- Submit for Simulation
- Submit for Approval
- Approve
- Reject
- Schedule
- Activate
- Pause
- Resume
- Suspend
- Extend
- Terminate
- Archive

Safety Controls

Before activation TICVAI shall verify:

- Required configuration is completed.
- Valid products exist.
- Dates are valid.
- Promotion rules do not conflict.
- Budget exists where required.
- Channels are assigned.
- Customer eligibility exists.
- Approval is completed.
- Promotion does not violate configured discount ceilings.

Emergency Suspension

Authorized users shall be able to immediately suspend a promotion if:

- Financial exposure is abnormal.
- Fraud is detected.
- Incorrect discount occurs.
- A partner requests suspension.
- Inventory becomes unavailable.
- Campaign budget is exhausted.

Suspension reason must be recorded.

Screen 5 — Campaign Calendar & Timeline

Purpose

Provide a calendar-based operational view of promotions.

Views

- Day
- Week
- Month
- Quarter
- Campaign timeline

Display

Show promotions by:

- Venue
- Product
- Attraction
- Channel
- Campaign
- Promotion family

Visual Status Indicators

Identify:

- Active
- Upcoming
- Ending soon
- Expired
- Pending approval
- Conflicting
- Suspended

Calendar Actions

Users may:

- Drag/change dates subject to permission.
- Open campaign details.

- Duplicate campaigns.
- Extend promotion periods.
- Compare overlapping campaigns.

Conflict Detection

The system shall flag situations such as:

Summer Family Promotion and Bank Partner Promotion both applying to the same product, segment, date, and channel.

The user can then open the Promotion Conflict Engine defined in a later board.

Screen 6 — Promotion Channel & Publication Monitor

Purpose

Ensure promotional configurations are correctly synchronized across every TICVAI sales channel.

Supported Channels

- B2C Website
- B2B Portal
- POS
- Mobile POS
- Kiosk
- Mobile App
- Call Center
- Guest Portal
- API
- OTA
- Reseller
- Partner Portal
- F&B POS
- Retail POS

Publication Status

For each channel display:

- Not Assigned
- Pending Publication
- Published
- Synchronizing
- Publication Failed
- Out of Sync
- Suspended

Channel Details

Show:

- Promotion version

- Last synchronized
- Rules published
- Products published
- Code availability
- Channel restrictions
- Error messages

Actions

- Publish
- Republish
- Disable channel
- Compare configurations
- View synchronization log

This is critical because promotion results must remain consistent across the booking platform and all consumer-facing channels.

Screen 7 — Promotion Alerts & Exception Center

Purpose

Centralize operational, commercial, and financial alerts affecting promotions.

Alert Categories

Configuration

- Missing product
- Missing eligibility
- Invalid dates
- Invalid discount
- Invalid code
- Missing approval

Financial

- Budget near limit
- Budget exceeded
- Margin below threshold
- Excessive discount exposure

Operational

- Promotion failed to publish
- Product unavailable
- Bundle component unavailable
- Channel synchronization failure

Commercial

- Low conversion
- Low redemption
- Unexpected high redemption

- Campaign underperforming

Fraud/Risk

- Abnormal coupon usage
- Excessive repeat redemption
- Suspicious customer behavior
- Promo-code leakage

Severity

- Information
- Warning
- Critical

Actions**Users can:**

- Assign alert
 - Acknowledge
 - Resolve
 - Suspend campaign
 - Open related configuration
 - Add internal comment
-

Screen 8 — Promotion Approval Inbox

Purpose

Provide one centralized approval workspace for promotional changes.

Approval Queue**Show:**

- Promotion
- Request type
- Requested by
- Requested date
- Discount exposure
- Revenue impact estimate
- Margin impact
- Campaign budget
- Risk level
- Requested activation date
- Current approval level

Approval Types**Approval can be required for:**

- New promotion
- Discount increase
- Budget increase

- Campaign extension
- Rule modification
- Product expansion
- Channel expansion
- Emergency promotion
- Promotion cancellation

Multi-Level Approval

Example:

Marketing Manager → Commercial Director → Finance → General Manager

Approval chains must be configurable.

Actions

- Approve
- Reject
- Return for Change
- Request Information
- Delegate

All decisions shall become part of the permanent audit trail.

This directly supports the matrix requirement for configurable multi-level approval and audit tracking for promotion creation, modification, activation, and deactivation.

Screen 9 — Promotion Health & Performance Monitor

Purpose

Provide near-real-time operational performance monitoring while campaigns are running.

KPIs

- Impressions
- Promotion views
- Eligible transactions
- Promotion applications
- Redemptions
- Redemption rate
- Conversion rate
- Gross sales
- Net sales
- Discount granted
- Incremental revenue
- AOV uplift
- Margin
- Cost per redemption
- Budget consumed
- Budget remaining

Comparison

Compare:

- Promotion vs baseline
- Promotion vs previous campaign
- Promotion vs AI forecast
- Promotion vs control group
- Channel vs channel
- Venue vs venue

Automated Rules

Example:

If margin impact exceeds -15%, generate Critical Warning.

If campaign budget reaches 90%, generate Budget Warning.

If conversion uplift exceeds target by 25%, AI may recommend extending campaign.

Detailed analytics and experimentation will exist later; this screen is primarily the **operational health monitor**.

Screen 10 — Promotion Audit, Activity & Version History

Purpose

Provide complete governance and traceability for every promotion.

Audit Information

Record:

- Promotion created
- Promotion edited
- Rule changed
- Discount changed
- Product added/removed
- Eligibility changed
- Channel changed
- Dates changed
- Budget changed
- Approval submitted
- Approval granted/rejected
- Promotion activated
- Promotion paused
- Promotion suspended
- Promotion expired
- Promotion archived

Audit Fields

For every change:

- User

- Role
- Date/time
- Previous value
- New value
- Reason
- Approval reference
- Promotion version

Version Comparison

Users can compare:

Version 2.3 ↔ Version 2.4

and see exactly what changed.

Restore / Clone

Subject to authorization:

- Restore previous configuration.
 - Clone previous version as new draft.
 - Create new promotion from historical configuration.
-

AI Requirements for Board 1

AI should operate as a **decision-support layer**, not automatically alter commercial configurations unless explicitly authorized.

AI shall:

- Rank campaigns requiring attention.
- Detect abnormal redemption.
- Detect unusual discount exposure.
- Detect promotion underperformance.
- Forecast budget exhaustion.
- Flag likely campaign conflicts.
- Identify margin deterioration.
- Recommend campaign extension.
- Recommend suspension.
- Recommend optimization.
- Identify potentially successful campaigns worth replicating.
- Generate natural-language summaries.

Example:

AI Campaign Summary

FAMILY30 is generating +19% conversion and +11% AOV, but discount exposure is 8% above forecast. At the current redemption rate the AED 250,000 campaign budget is projected to reach its limit three days before campaign end.

Roles & Permissions

Board 1 should support granular PBAC/RBAC.

Typical roles:

- Marketing Administrator
- Campaign Manager
- Commercial Manager
- Revenue Manager
- Finance
- Venue Manager
- Operations
- B2B Manager
- System Administrator
- Approver
- Auditor
- Read Only

Permissions should differentiate:

- View
- Create
- Edit
- Submit
- Approve
- Activate
- Pause
- Suspend
- Change budget
- Archive
- Export

Key Integrations

Board 1 must integrate with:

Ticketing Catalogue

Products, ticket types, variants and availability.

Pricing / Dynamic Pricing

Base prices and pricing strategies.

CRM

Segments, customer profiles and behavior.

Membership & Loyalty

Membership eligibility, benefits and loyalty rewards.

B2B / Reseller

Partner/customer hierarchy, special pricing and commissions.

F&B and Retail

Cross-category promotion products.

Wallet / Voucher / Gift Card

Eligible value instruments.

Payment Orchestration

Payment-method and card-partner offers.

Finance

Revenue, discount cost, allocation and profitability.

Reporting / AI Analytics

Campaign analytics and AI insights.

Matrix Coverage — Board 1

Board 1 primarily covers the **management, governance and operational supervision** of the promotion requirements rather than individual promotion mechanics.

It directly contributes to requirements including:

- **3.6.12** — Back-office promotion management.
- **3.6.35** — Promotion hierarchy/conflict visibility.
- **3.6.36** — Campaign budgets, limits and automated suspension.
- **3.6.38** — Approval workflows and audit tracking.
- **3.6.39** — Campaign variant status/visibility.
- **3.6.40** — AI promotion recommendations.
- **4.1.6** — Promotion engine with multiple draft promotion options.
- **4.1.18** — Automatic promotion recommendations.
- **4.1.20** — Promotion strategy recommendations.
- **4.2.1** — Visibility into promotion combinations/conflicts.
- **4.2.5** — Promotion dimensions including location/channel/group/type.

The detailed mechanics behind these requirements will subsequently be configured through Boards 2–10.

Acceptance Criteria — Board 1**Board 1 shall be considered complete when:**

1. Authorized users can view every promotion from one centralized command center.
2. Promotions can be filtered by status, type, product, campaign, venue, channel, segment, owner and date.
3. Users can navigate from the command center into every related promotion configuration area.
4. Promotion lifecycle status can be controlled through governed actions.
5. Promotion activation cannot bypass mandatory validations and approval rules.
6. Authorized users can immediately suspend an active promotion.
7. Promotion budgets, financial exposure and operating health can be monitored.
8. Promotion publication status is visible across all configured sales channels.
9. Campaign operational alerts are automatically generated.
10. Approval requests can be reviewed from one centralized inbox.
11. Promotion performance can be monitored during campaign execution.
12. Complete activity and configuration history is retained.
13. Every change identifies the responsible user, timestamp and changed values.
14. AI generates alerts, recommendations and campaign summaries.

15. AI recommendations require authorized human approval before changing commercial rules.
16. The system supports multi-tenant, multi-venue and multi-business-entity operation.
17. Permissions follow TICVAI RBAC/PBAC governance.
18. All screens support auditability and version control.
19. Promotion information stays synchronized with downstream sales channels.
20. The architecture supports future promotion types without requiring redesign of the command center.

Board 1 is therefore locked as: *Promotion Command Center — 10 backend screens.*

The next logical board is **Board 2 — Promotion & Discount Rule Studio**, which will be one of the most important boards because that is where we define the actual percentage discounts, fixed discounts, bulk pricing, early bird, last minute, cart thresholds, guest discounts and other promotion mechanics from the matrix.

Module 1 — Promotions & Bundles Management

Board 2 — Promotion & Discount Rule Studio

Board Objective

Board 2 is the **core commercial rule engine** for promotions.

It shall allow authorized TICVAI business users to configure discount and promotion mechanics without development, including percentage and fixed discounts, transaction thresholds, volume discounts, bulk pricing, early-bird and last-minute offers, special guest pricing, payment-method discounts, partner discounts, and dynamic conditional discounts.

The matrix explicitly requires percentage/value discounts, quantity thresholds, bulk discounts, early-bird/volume/customer-segment conditions, payment-type offers, group pricing, and special individual pricing.

The board should contain **10 backend screens**.

Screen 1 — Promotion Rule Builder

Purpose

Provide the main no-code workspace for creating the commercial logic behind a promotion.

Rule Structure

TICVAI should use a simple:

IF → THEN → AND/OR → EXCEPT

logic.

Example:

IF Ticket Type = General Admission
AND Quantity ≥ 4

AND Sales Channel = B2C
AND Purchase Date before 15 December
THEN Apply 15% Discount
EXCEPT Annual Pass Members.

Configuration Areas

Rule Information

- Rule name
- Rule ID
- Promotion
- Description
- Owner
- Business entity
- Venue
- Rule status
- Priority

IF Conditions

- Product
- Ticket type
- Product category
- Quantity
- Basket amount
- Guest type
- Customer segment
- Membership level
- Loyalty tier
- Channel
- Venue
- Attraction
- Location
- Day/date
- Time
- Payment method
- Partner
- Promo code

THEN Actions

- Percentage discount
- Fixed discount
- Fixed selling price
- Free product
- Free ticket
- Upgrade

- Added value
- Voucher
- Reward entitlement

Logic Controls

Support:

- AND
 - OR
 - NOT
 - Nested condition groups
 - Multiple outcomes
 - Rule ordering
-

Screen 2 — Percentage & Fixed Discount Configurator

Purpose

Configure the two fundamental discount types required by the matrix.

The matrix explicitly states that discounts may be defined as **percentage or fixed value**.

Discount Types

Percentage

Example:

15% off General Admission.

Configure:

- Discount percentage
- Maximum percentage
- Maximum monetary discount
- Minimum qualifying amount
- Rounding method

Fixed Value

Example:

AED 50 off purchases above AED 500.

Configure:

- Discount amount
- Currency
- Minimum basket value
- Maximum uses
- Whether applied per item or transaction

Fixed Promotional Price

Example:

Standard price AED 150 → Promotional price AED 119.

Price Basis

Allow the discount to apply against:

- Standard price
- Current selling price
- Dynamic price
- Membership price
- B2B price
- Package price

Protection Rules

Prevent:

- Negative prices
 - Price below configured floor
 - Discount above authorized ceiling
 - Margin below minimum threshold
-

Screen 3 — Cart & Transaction Threshold Rules

Purpose

Configure promotions triggered by basket value, ticket quantity, transaction value, or purchase composition.

The matrix specifically requires rules such as **if more than X tickets are purchased, apply Y discount to the entire transaction.**

Threshold Types

- Basket value $\geq X$
- Quantity $\geq X$
- Number of specific products $\geq X$
- Number of ticket products $\geq X$
- Number of categories $\geq X$
- Total spend within category $\geq X$

Examples

Spend AED 500 → receive 10% off.

Buy 5 tickets → receive AED 100 discount.

Buy 4 attraction tickets → entire transaction receives 15%.

Threshold Bands

Support:

Spend	Discount
AED 250	5%
AED 500	10%
AED 1,000	15%

Qualification Logic

Configure whether:

- Tax counts toward threshold.
 - Fees count.
 - Vouchers count.
 - Discounts are evaluated before/after threshold.
 - Voided items are excluded.
 - Refunded items affect qualification.
-

Screen 4 — Volume, Bulk & Tier Discount Configurator**Purpose**

Manage quantity-based and bulk-purchase commercial rules.

The matrix requires configurable bulk thresholds and discount percentages, dedicated group pricing, and tiered bulk purchasing.

Volume Tiers**Example:**

Quantity	Discount
1-4	0%
5-9	5%
10-14	10%
15-30	15%
31+	20%

Configuration

- Minimum quantity
- Maximum quantity
- Discount %
- Discount value
- Fixed unit price
- Eligible product
- Eligible customer
- Eligible channel

Group Pricing**Support scenarios such as:**

Groups of 15+ receive dedicated group rates.

B2B Application**Allow different tiers for:**

- Schools

- Tour operators
- Travel agencies
- Resellers
- Corporate accounts
- Government entities

Voucher Pack Support

Bulk purchases may trigger generation of:

- Individual ticket pack
- Voucher pack
- PDF ticket pack

as required by the matrix's bulk-purchase configuration.

Screen 5 — Time-Based & Seasonal Discount Rules

Purpose

Configure promotional pricing based on when the customer purchases or visits.

Promotion Types

Early Bird

Example:

Purchase ≥ 30 days before visit → 15% discount.

Last Minute

Example:

Purchase within 2 hours of timeslot → 20% discount.

Seasonal

- Summer
- Ramadan
- Eid
- School holidays
- National Day
- Peak/off-peak
- Custom seasons

Flash Sale

Example:

25% discount today from 18:00–22:00.

Conditions

- Purchase date
- Visit date
- Days-before-visit
- Hours-before-visit
- Day of week
- Time

- Timeslot
- Season
- Event period

Flash-sale requirements and promotion validity windows are explicitly included in the matrix.

Screen 6 — Customer, Membership & Segment Discount Rules

Purpose

Configure discounts based on who the customer is.

Eligibility Dimensions

- Customer segment
- CRM segment
- Guest category
- Age category
- Membership status
- Membership tier
- Loyalty tier
- Annual pass holder
- Corporate affiliation
- Partner affiliation
- Account
- Country/residency
- B2B customer

Example Segments

- Annual Pass Holder
- Fazaa
- UAE Pass-linked segment
- Corporate Employee
- VIP
- School
- Travel Agency
- Reseller
- Family
- Senior
- Child

The provided matrix specifically references segment discounts such as annual pass holders and account-target discounts.

Account-Specific Pricing

Example:

Aldar employee account → 20% General Admission discount.

Privacy Principle

The rule should reference a **segment or eligibility attribute**, rather than exposing unnecessary customer data to promotion administrators.

Screen 7 — Payment Method, Bank & Partner Discount Rules

Purpose

Configure discounts triggered by how the guest pays or which commercial partner they belong to.

The matrix specifically includes discounts for payment types and bank credit/debit cards.

Payment Conditions

- Credit card
- Debit card
- Visa
- Mastercard
- Mada
- Apple Pay
- Wallet
- Gift card
- Bank-specific card
- Selected payment gateway

Example

Pay using eligible VISA card → 10% discount.

Bank Campaign

Configure:

- Partner bank
- BIN/IIN eligibility reference
- Promotion period
- Eligible products
- Minimum spend
- Discount %
- Maximum discount
- Number of uses
- Customer limit
- Campaign budget

Partner Offers

Support:

- Banks
- Hotels
- Airlines
- Tourism partners
- Corporate partners

- Government partners
- Membership programs

Actual card validation should be performed through the **Payment Orchestration integration**, not by storing raw card information in Promotions.

Screen 8 — Special Price & Guest Offer Configurator

Purpose

Manage commercially distinct special-price products and targeted offers without unnecessarily duplicating product SKUs.

Special Pricing Examples

The matrix specifically calls out:

- Two-parks pass
- Multi-parks pass
- Ladies night
- Annual pass
- Family pass

Configuration

- Offer name
- Eligible product
- Eligible guest
- Price
- Discount
- Valid dates
- Valid visit dates
- Quantity
- Channel
- Venue
- Capacity
- Restrictions

Additional Examples

- Resident price
- Tourist price
- Employee price
- Student price
- School price
- Family price
- Group price
- Partner price

Product Architecture

Where possible, offers should reference the existing ticket/product definition rather than create duplicated SKUs, consistent with the matrix's attribute-driven product approach.

Screen 9 — Discount Limits, Guardrails & Commercial Controls

Purpose

Protect the business from incorrectly configured discounts and excessive commercial exposure.

Guardrails

Configure:

- Maximum discount %
- Maximum discount value
- Minimum selling price
- Minimum margin
- Maximum transaction discount
- Maximum customer discount
- Maximum campaign exposure
- Maximum redemption count
- Per-customer usage
- Per-account usage

Override Policy

Different authority levels may be allowed to exceed limits.

Example:

Marketing Executive

- Maximum discount: 15%

Marketing Manager

- Maximum discount: 25%

Commercial Director

- Maximum discount: 40%

CEO/CFO approval

- Above 40%

Runtime Actions

When limits are breached:

- Reject
- Require approval
- Generate warning
- Suspend rule
- Suspend promotion

Campaign budgets, redemption limits, discount caps and automatic suspension are explicitly required by the matrix.

Screen 10 — Rule Test, Simulation & AI Recommendation Workspace

Purpose

Allow administrators to test promotional rules before activating them.

This is critical because the matrix requires simulation of redemption, discount exposure, revenue impact, margin impact and financial performance before activation.

Test Basket

Administrators can create a sample transaction:

- Guest
- Segment
- Products
- Quantity
- Channel
- Date
- Venue
- Membership
- Loyalty
- Payment type
- Promo code

Simulation Result

Show:

Original basket

AED 600

Applied promotion

Family Volume Discount

Discount

15% / AED 90

Final amount

AED 510

Applied rules

- Quantity ≥ 4 ✓
- Family segment ✓
- B2C ✓
- Valid date ✓

Rule Trace

Display exactly:

Rule 421 qualified.

Rule 330 failed because quantity < 5 .

Rule 190 excluded because customer is not a member.

This will be extremely valuable for operational support.

AI Requirements for Board 2

AI should assist with **commercial rule creation and optimization**, while humans retain approval authority.

AI shall be able to:

- Recommend promotion mechanics.
- Recommend percentage/value levels.
- Identify unnecessarily complex rules.
- Suggest qualifying thresholds.
- Recommend early-bird windows.
- Recommend bulk tiers.
- Recommend target segments.
- Predict redemption.
- Estimate revenue impact.
- Estimate margin impact.
- Identify potentially overlapping rules.
- Detect rules likely to cannibalize standard pricing.
- Suggest safer discount caps.
- Explain configured rules in plain language.

Example:

AI Recommendation:

Historical family-ticket transactions show the strongest conversion uplift at 4+ tickets.

Consider changing the threshold from 5 tickets to 4 and reducing the discount from 20% to 15%. Estimated incremental revenue: +AED 180K while preserving margin.

The matrix explicitly calls for AI recommendations for promotion structure, discount percentage, validity, audience and campaign timing.

Roles & Governance

Support roles including:

- Marketing Administrator
- Campaign Manager
- Commercial Manager
- Revenue Manager
- Finance
- B2B Manager
- Venue Manager
- System Administrator
- Approver
- Auditor

Permissions must independently control:

- Create rules
- Edit rules
- Change discount
- Change thresholds

- Change segments
 - Change dates
 - Override limits
 - Run simulation
 - Submit
 - Approve
 - Activate
-

Key Integrations

Ticketing Catalogue

Products, ticket types, variants and base price.

Pricing / Dynamic Pricing

Current price and pricing floors.

CRM

Guest/customer segments.

Membership & Loyalty

Membership and tier eligibility.

B2B / Reseller

Company segments and partner pricing.

Payment Orchestration

Payment-method and bank eligibility.

F&B / Retail

Products eligible for promotional rules.

Finance

Revenue, margin and discount exposure.

AI / Analytics

Historical performance and recommendations.

Board 2 Matrix Coverage

This board primarily covers or contributes to:

3.6.5 — quantity-triggered transaction discounts.

3.6.6 — configurable bulk discounts.

3.6.8 — payment/partner discount offers.

3.6.9-3.6.10 — flash-sale mechanics.

3.6.14 — percentage/fixed-value basket discounts.

3.6.18-3.6.19 — payment method and bank/card discounts.

3.6.22-3.6.25 — discount eligibility, percentage/value configuration and free/discounted offers.

3.6.27-3.6.28 — controlled discount access and group pricing.

3.6.30-3.6.33 — special prices, dynamic discount conditions and bulk tiers.

Board 2 also prepares the rule layer for later boards handling **BOGO, coupons, stacking/conflicts, targeting, campaign budgets and AI optimization**, so we avoid duplicating those functions.

Acceptance Criteria

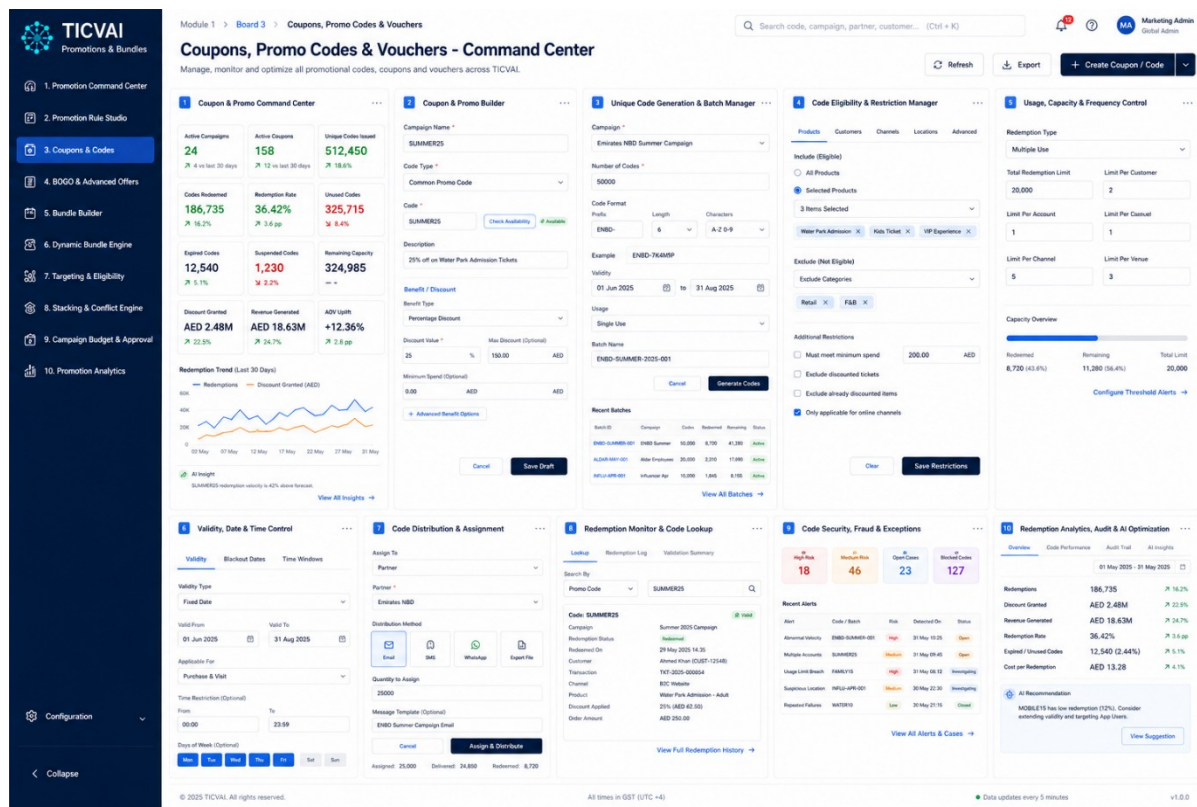
Board 2 shall be complete when:

1. Authorized users can create promotion rules without software development.
2. Percentage and fixed-value discounts are supported.
3. Fixed promotional prices are supported.
4. Cart-value and quantity thresholds are configurable.
5. Multiple threshold tiers are supported.
6. Bulk/group prices can be configured.
7. Early-bird and last-minute conditions are supported.
8. Seasonal and flash-sale conditions are supported.
9. Customer/CRM segments can determine eligibility.
10. Membership and loyalty status can determine eligibility.
11. B2B/partner segments can determine eligibility.
12. Payment method and eligible bank/card programs can trigger promotions.
13. Special guest and product pricing can be configured.
14. Rules support AND/OR/NOT and nested conditions.
15. Discount floors, caps and margin guardrails can be enforced.
16. Unauthorized users cannot exceed their permitted commercial limits.
17. Users can test a rule against a simulated basket.
18. The system explains why each rule qualified or failed.
19. Financial impact can be simulated before activation.
20. AI can recommend rule structure and discount values.
21. AI cannot activate or modify commercial rules without the configured authorization.
22. Every configuration change is versioned and audited.
23. Rules are reusable across eligible products, venues and channels.
24. The architecture supports additional rule conditions without redesigning the promotion engine.

Board 2 is therefore locked as: *Promotion & Discount Rule Studio — 10 backend screens.*

Module 1 — Promotions & Bundles Management

Board 3 — Coupons, Promo Codes & Vouchers



Board Objective

Board 3 shall provide TICVAI with a centralized **Coupon, Promo Code & Promotional Voucher Management Engine** for creating, generating, distributing, validating, redeeming, tracking, suspending, and auditing promotional codes across all eligible sales channels.

This board covers the matrix requirements around common and unique promo codes, single/multiple usage, date/time restrictions, capacity and frequency limits, location/group/channel restrictions, coupon discounts, redemption reporting, free or discounted offers, and bulk voucher generation.

The board shall contain **10 backend screens**.

Screen 1 — Coupon & Promo Code Command Center

Purpose

Provide the central operational dashboard for all coupon, promo-code, and promotional voucher activities.

KPI Cards

- Active Code Campaigns
- Active Coupons
- Unique Codes Issued
- Codes Redeemed
- Redemption Rate

- Unused Codes
- Expired Codes
- Suspended Codes
- Remaining Redemption Capacity
- Discount Granted
- Revenue Generated
- Average Order Value
- Fraud/Suspicious Usage Alerts

Campaign Status

Support:

- Draft
- Pending Approval
- Scheduled
- Active
- Paused
- Suspended
- Capacity Reached
- Expired
- Cancelled
- Archived

Performance Panels

Show:

- Top-performing codes
- Redemption trend
- Redemption by channel
- Redemption by venue
- Redemption by partner
- Redemption by customer segment
- Discount exposure
- Campaign budget consumption

AI Insights

Examples:

AI Alert: SUMMER25 redemption velocity is 42% above forecast.

AI Recommendation: PARTNER15 has strong conversion but low distribution penetration. Consider expanding distribution to the Mobile App.

Screen 2 — Coupon & Promo Code Builder

Purpose

Create the commercial definition of a coupon or promo-code campaign.

Code Type

Users select:

- Common promo code
- Unique promo code
- Coupon
- Promotional voucher
- Free-ticket code
- Discount voucher
- Partner code
- Employee code
- Influencer/affiliate code
- Compensation/service-recovery code
- Bulk campaign code

Example**Campaign:** SUMMER25**Code:** SUMMER25**Reward:** 25% discount**Maximum discount:** AED 150**Eligible product:** Water Park Admission**Channel:** B2C + Mobile App**Validity:** 1 June–31 August**Commercial Benefit****Configure:**

- Percentage discount
- Fixed-value discount
- Fixed promotional price
- Free product
- Free ticket
- Free add-on
- Upgrade
- Bundle benefit
- Added value

The actual discount calculation shall consume the **Board 2 Promotion Rule Engine**, rather than duplicate discount logic.

Screen 3 — Unique Code Generation & Batch Manager

Purpose

Generate and manage large quantities of secure unique promotional codes.

Generation Configuration**Users define:**

- Campaign
- Number of codes
- Code length

- Prefix
- Suffix
- Character type
- Case sensitivity
- Expiration
- Number of uses
- Distribution owner

Example:

Generate 50,000 unique codes for Emirates NBD Summer Campaign.

Code Format Example

ENBD-7K4M9P

ENBD-X8T2QA

Batch Information

Each batch records:

- Batch ID
- Campaign
- Quantity generated
- Generated by
- Generation date
- Expiry
- Assigned partner
- Distribution status
- Redeemed quantity
- Remaining quantity

Security

Codes should be generated using sufficiently unpredictable values where fraud risk requires it.

Administrators should not need access to the entire raw code inventory unless their permissions explicitly allow it.

Screen 4 — Code Eligibility & Restriction Manager

Purpose

Determine **where, when, by whom, and against what** a code can be redeemed.

The matrix explicitly requires promo codes to support restrictions for usage, dates, duration, capacity, frequency, location, group, partner, operating area, and sales channel.

Product Restrictions

- Ticket
- Ticket type
- Product
- Product category
- Attraction

- Event
- Bundle
- Membership
- F&B
- Retail
- Add-on

Customer Restrictions

- Guest type
- CRM segment
- Membership
- Loyalty tier
- B2B account
- Partner
- Corporate group

Channel Restrictions

- B2C
- B2B
- POS
- Mobile POS
- Kiosk
- Mobile App
- Call Center
- Partner
- Reseller
- API

Geographic Restrictions

- Business entity
- Venue
- Attraction
- Location
- Operating area

Rule Logic**Support:****INCLUDE**

and

EXCLUDE

conditions.

Screen 5 — Usage, Capacity & Frequency Control

Purpose

Control exactly how frequently and how many times promotional codes may be redeemed.

Usage Configuration

Support:

- Single use
- Multiple use
- Unlimited use
- Maximum total redemptions
- Maximum per customer
- Maximum per account
- Maximum per transaction
- Maximum per day
- Maximum per channel
- Maximum per venue

Frequency Controls

Examples:

One use per booking.

One redemption per customer every 30 days.

Maximum 5 uses per corporate account.

Campaign maximum 20,000 redemptions.

Capacity Counter

Show:

Issued: 50,000

Redeemed: 31,450

Reserved/Pending: 420

Remaining: 18,130

Threshold Actions

At:

- 80% → informational alert
- 90% → warning
- 100% → automatically stop redemption

Thresholds shall be configurable.

Screen 6 — Validity, Date & Time Control

Purpose

Control the temporal validity of coupons and codes.

Validity Types

Fixed Date

1 June 2027 → 31 August 2027

Relative Validity

Valid for 30 days after issuance.

Purchase Window

Must purchase before 15 December.

Visit Window

Ticket must be used between 1–30 January.

Time Specific

Valid 18:00–22:00.

Day Specific

Monday–Thursday only.

Additional Controls

- Blackout dates
- Holidays
- Selected timeslots
- Selected events
- Seasonal calendars
- Expiration grace period

Timezone

All validity calculations must respect the applicable **venue/business timezone**, not simply the administrator's browser timezone.

Screen 7 — Code Distribution & Assignment Manager

Purpose

Manage how promotional codes are allocated and distributed.

Distribution Channels

- Email
- SMS
- WhatsApp
- Mobile App
- CRM journey
- Guest portal
- B2B portal
- Partner portal
- POS
- Call center
- API
- Exported batch

Assignment**Codes may be assigned to:**

- Individual customer

- Customer segment
- Membership account
- B2B company
- Reseller
- Travel agency
- School
- Hotel
- Bank
- Corporate partner
- Marketing campaign

Distribution Tracking

Show:

- Generated
- Assigned
- Sent
- Delivered
- Viewed where available
- Redeemed
- Expired
- Cancelled

External Partner Example

Partner: Bank ABC

Batch: BANKABC-2027-001

Codes: 25,000

Assigned: 25,000

Redeemed: 8,720

Remaining: 16,280

Actual outbound communications should integrate with TICVAI's CRM/Marketing services rather than duplicating a complete communication engine here.

Screen 8 — Redemption Monitor & Code Lookup

Purpose

Provide real-time operational visibility into coupon and promo-code redemption.

Search

Authorized users can search by:

- Promo code
- Coupon ID
- Batch ID
- Transaction
- Booking
- Customer

- Partner
- Campaign

Redemption Record

Show:

- Code
- Campaign
- Redemption date/time
- Transaction
- Booking
- Customer/account reference
- Product
- Original value
- Discount
- Final value
- Channel
- Venue
- Device/POS
- Operator
- Validation result

Validation Results

- Valid
- Redeemed
- Expired
- Not Started
- Usage Limit Reached
- Invalid Product
- Invalid Channel
- Invalid Customer
- Invalid Location
- Suspended
- Cancelled

The matrix explicitly requires visibility into whether a coupon has been redeemed and requires redemption reporting.

Screen 9 — Code Security, Fraud & Exception Center

Purpose

Detect promo-code abuse, leakage, abnormal redemption, and suspicious campaign behavior.

Detection Rules

Monitor:

- Excessive redemption velocity
- Repeated failed attempts
- Multiple customers using customer-specific code
- Unusual geographic usage
- High-volume redemption from one device
- Suspicious POS/operator activity
- Code enumeration attempts
- Partner code leakage
- Redemption above expected campaign pattern

Risk Levels

- Low
- Medium
- High
- Critical

Actions

Authorized users can:

- Suspend individual code
- Suspend batch
- Suspend campaign
- Block redemption
- Reinstate code
- Assign investigation
- Add case note

AI Fraud Detection

AI can identify unusual patterns such as:

A partner campaign normally generates 100–150 redemptions/hour. Redemption increased to 1,800/hour from a concentrated device/IP pattern.

AI should flag the anomaly; governed actions remain subject to TICVAI's configured authorization policies.

Screen 10 — Redemption Analytics, Audit & AI Optimization

Purpose

Provide complete performance analytics and governance for coupon and promo-code campaigns.

Performance KPIs

- Codes generated
- Codes distributed
- Codes redeemed
- Redemption rate
- Conversion rate

- Revenue generated
- Discount granted
- Incremental revenue
- AOV uplift
- Cost per redemption
- Margin impact
- Expired unused codes

Analysis Dimensions

Analyze by:

- Campaign
- Code
- Batch
- Product
- Venue
- Channel
- Customer segment
- Partner
- Date/time
- Promotion type

Funnel

Generated → Distributed → Eligible → Applied → Redeemed → Revenue

AI Recommendations

AI may recommend:

- Increase/decrease discount
- Extend validity
- Reduce validity
- Increase code inventory
- Suspend low-performing codes
- Expand channels
- Target another segment
- Change usage limit
- Replicate successful campaign

Audit History

Record:

- Created
- Modified
- Generated
- Assigned
- Distributed
- Redeemed

- Suspended
- Reactivated
- Expired
- Cancelled

Including:

- User
 - Timestamp
 - Previous value
 - New value
 - Reason
 - Approval reference
-

Important Architecture Principle

Board 3 must **not create another promotion engine**.

The architecture should work as:

Coupon / Promo Code

↓ identifies benefit

Board 2 — Promotion Rule Engine

↓ determines qualification and calculation

Board 8 — Stacking & Conflict Engine

↓ determines interaction with other offers

Pricing Engine

↓

Shopping Cart / Transaction

This keeps one authoritative commercial calculation engine across TICVAI.

AI Requirements for Board 3

AI shall support:

- Redemption forecasting
- Abnormal redemption detection
- Promo-code leakage detection
- Campaign performance prediction
- Code quantity recommendation
- Optimal validity recommendation
- Target segment recommendation
- Channel recommendation
- Discount optimization
- Campaign extension recommendation

- Partner performance analysis

AI should explain recommendations in business language.

Example:

AI Recommendation: Only 21% of MOBILE20 codes have been redeemed, but customers receiving the code through the mobile application show a 31% conversion rate. Consider reallocating unused codes from email to mobile-app audiences.

AI recommendations must not independently alter financial rules unless explicitly permitted by governed automation policies.

Key Integrations

Promotion Rule Engine — discount calculation and qualification.

Ticketing Catalogue — eligible tickets/products.

Bundles — bundle-related coupon benefits.

CRM & Marketing — customer assignment and campaign distribution.

Membership & Loyalty — member eligibility.

B2B / Reseller — partner and reseller campaigns.

F&B / Retail — cross-category redemption.

POS / Sales Channels — code validation and redemption.

Payment — payment-method-dependent promotions.

Finance — discount exposure and revenue impact.

Reporting / AI — analytics and optimization.

Matrix Coverage — Board 3

Board 3 directly addresses major portions of:

- **3.6.9-3.6.10** — promo-code-based flash campaigns.
- **3.6.11** — once-per-booking restriction.
- **3.6.13** — single/multiple use, common/unique codes, dates, duration, capacity, frequency, location, group, partner, operating area and channel restrictions.
- **3.6.14** — coupon-based percentage/value discounts.
- **3.6.15** — coupon redemption status/report.
- **3.6.23** — promo codes for free/discounted offers.
- **3.6.24** — unique/common codes.
- **3.6.25** — percentage or defined-value benefit.
- **3.6.33** — bulk-purchase voucher/PDF packs.

Related discount calculations remain governed by **Board 2**, while stacking/conflict behavior will be governed by a later board.

Acceptance Criteria

Board 3 shall be considered complete when:

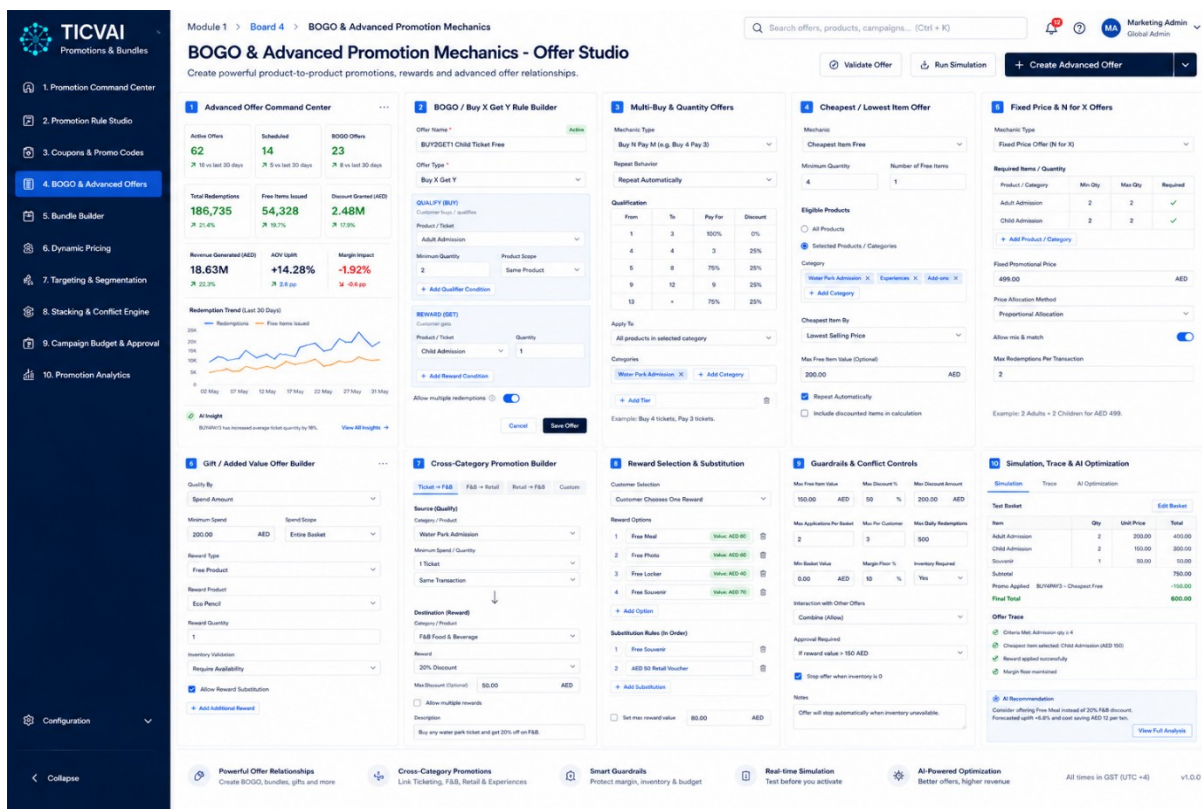
1. Users can create common and unique promotional codes.
2. Coupons and promotional vouchers can be configured.
3. Large unique-code batches can be generated.
4. Codes can support single, multiple, capped, or unlimited redemption.

5. Usage can be limited per booking, customer, account, channel, venue, or campaign.
6. Date/time/relative validity rules are supported.
7. Product and ticket eligibility can be configured.
8. Customer and segment eligibility can be configured.
9. Channel, partner, and location restrictions can be configured.
10. Codes can trigger percentage, value, free-product, free-ticket, and other configured benefits.
11. Codes can be allocated to partners and customers.
12. Distribution status can be tracked.
13. Redemption can be validated in real time.
14. Users can determine exactly why a code was accepted or rejected.
15. Redemption reporting is available.
16. Codes, batches, or campaigns can be suspended.
17. Suspicious redemption patterns generate alerts.
18. Campaign performance and discount exposure can be measured.
19. AI can recommend campaign optimizations.
20. Complete code lifecycle and configuration activity is audited.
21. Promo-code calculation reuses TICVAI's central promotion rule engine.
22. RBAC/PBAC controls protect code generation, export, suspension, and financial configuration.
23. The design supports multi-tenant, multi-venue, multi-channel, and multi-currency operations.
24. New code types can be introduced without redesigning the underlying promotion architecture.

Board 3 is locked as: *Coupons, Promo Codes & Vouchers — 10 backend screens.*

Module 1 — Promotions & Bundles Management

Board 4 — BOGO & Advanced Promotion Mechanics



Board Objective

Board 4 shall provide TICVAI with an enterprise-grade **Advanced Promotion Mechanics Engine** for promotions involving relationships between products, quantities, basket composition, rewards, and qualifying purchases.

While **Board 2** defines standard discounts and thresholds and **Board 3** manages promo codes/coupons, Board 4 answers:

“When the customer buys X, what exactly should TICVAI give them, discount, replace, upgrade, or add to the transaction?”

The matrix requires mechanics such as Buy X Get X, Buy X Get Y, Buy N Get X, percentage/amount discounts on another product, cheapest-item-free, fixed-price combinations, cross-category F&B/Retail rewards, added-value gifts, and automatic cart-level promotion application.

Board 4 shall contain **10 backend screens**.

Screen 1 — Advanced Offer Command Center

Purpose

Provide the centralized management workspace for all advanced promotional mechanics.

KPI Cards

- Active Advanced Offers
- Scheduled Offers

- BOGO Campaigns
- Gift-with-Purchase Offers
- Fixed-Price Offers
- Cross-Category Offers
- Total Redemptions
- Free Items Issued
- Discount Granted
- Revenue Generated
- AOV Uplift
- Margin Impact

Offer Categories

Filter by:

- Buy X Get X
- Buy X Get Y
- Buy N Get X
- Buy N Get Multiple
- Cheapest Item Free
- Percentage Off Another Product
- Amount Off Another Product
- Fixed Bundle Price
- Gift with Purchase
- Added Value
- Cross-category reward
- Upgrade offer

Status

- Draft
- Pending Validation
- Pending Approval
- Scheduled
- Active
- Paused
- Suspended
- Expired
- Archived

AI Insights

Examples:

High Performer: BUY4PAY3 has increased average ticket quantity by 18%.

Margin Alert: RETAIL2FNB promotion is generating 11% more redemptions than forecast.

Screen 2 — Buy X Get Y / BOGO Rule Builder

Purpose

Configure the fundamental **qualifier** → **reward** relationship.

Rule Structure

BUY / QUALIFY

↓

GET / REWARD

Example:

Buy 2 Adult Admission Tickets

→ Get 1 Child Admission Ticket Free.

Qualifier Configuration

Configure:

- Product
- Product category
- Ticket type
- Quantity
- Minimum spend
- Customer segment
- Channel
- Venue
- Date/time

Reward Configuration

Configure:

- Same product
- Different product
- Product category
- Quantity
- Free
- Percentage discount
- Fixed discount
- Fixed reward price

Supported Mechanics

- Buy X → Get X free
- Buy X → Get Y free
- Buy N → Get X free
- Buy N → Get N free
- Buy X → Get Y discounted
- Buy N → Get X discounted

These mechanics are specifically required by the matrix.

Screen 3 — Multi-Buy & Quantity Offer Configurator

Purpose

Configure advanced quantity relationships that go beyond simple BOGO.

Examples

Buy 4 → Pay for 3.

Buy 6 → Pay for 5.

Buy 10 tickets → Get 2 tickets free.

Buy 3 F&B products → Get 1 selected item free.

Configuration

Qualifying Quantity

- Minimum quantity
- Exact quantity
- Quantity range
- Multiples of X

Reward Quantity

- One free
- Multiple free
- Percentage off
- Amount off
- Fixed total price

Repeat Behavior

Example:

Buy 4 Pay 3

If customer purchases 8:

- Apply once → Pay 7

or

- Repeat twice → Pay 6

Administrator shall configure:

- Apply once
- Repeat automatically
- Maximum repetitions

Mixed Product Qualification

Allow:

Buy any 4 products from selected category.

rather than requiring four identical SKUs.

Screen 4 — Cheapest / Lowest-Value Item Promotion

Purpose

Configure offers where TICVAI dynamically identifies the lowest-priced qualifying item.

The matrix explicitly requires “**buy multiple products and get cheapest item free**” and adding cheaper qualifying items within the same transaction.

Example

Customer buys:

Product	Price
Adult Ticket	AED 200
Adult Ticket	AED 200
Child Ticket	AED 150
Experience	AED 100

Promotion:

Buy 4 → Cheapest Free

TICVAI automatically discounts:

Experience = AED 100

Configuration

- Eligible products
- Eligible categories
- Minimum quantity
- Number of free items
- Cheapest/lowest-priced selection
- Maximum free-item value
- Maximum repetitions

Alternative Logic

Support:

- Cheapest item free
- Cheapest item 50% off
- N cheapest items free
- Cheapest qualifying item capped at X value

Screen 5 — Fixed-Price & “N for X” Offer Builder

Purpose

Configure promotions where a qualifying collection of products is sold for a fixed promotional total.

Examples

Any 3 tickets for AED 300.

4 attraction tickets for AED 450.

Family admission: 2 Adults + 2 Children = AED 499.

Configuration

- Required quantity
- Required products
- Product category
- Mix-and-match allowed
- Fixed promotional price
- Currency
- Maximum repetitions
- Minimum/maximum product values

Price Allocation

TICVAI shall define how the fixed promotional price is allocated across the qualifying items for:

- Refunds
- Tax
- Revenue recognition
- Partner settlement
- Reporting

This is important because the customer may see **AED 499**, but Finance still needs a defensible allocation across the underlying products.

Screen 6 — Gift, Free Product & Added-Value Offer Builder

Purpose

Configure promotions where a purchase generates an additional entitlement rather than simply reducing price.

The matrix explicitly provides the example:

“Buy for more than 200 AED and get a free pencil.”

Qualifying Conditions

- Basket value
- Product
- Quantity
- Product category
- Ticket
- Membership
- Customer segment

Reward Types

- Free retail product
- Free F&B product
- Free ticket
- Free add-on

- Free experience
- Voucher
- Upgrade
- Service
- Additional entitlement

Example

Spend AED 500 on attraction tickets
→ receive free souvenir.

Inventory Validation

For physical rewards, TICVAI must check:

- Inventory availability
- Location inventory
- Eligible fulfillment point
- Substitution rules

Out-of-Stock Behavior

Configure:

- Do not offer promotion
 - Provide alternative reward
 - Issue voucher
 - Allow later fulfillment
 - Escalate to operator
-

Screen 7 — Cross-Category Promotion Builder

Purpose

Create promotions spanning different TICVAI commercial domains.

The matrix explicitly requires:

Buy from Retail → receive something free from F&B, **and vice versa**.

Supported Domains

- Ticketing
- Attractions
- Events
- F&B
- Retail
- Membership
- Experiences
- Add-ons
- Parking
- Services

Examples

Ticket → F&B

Buy Water Park admission → Get 20% off lunch.

Retail → F&B

Spend AED 250 in Retail → Get free coffee.

F&B → Retail

Spend AED 150 at restaurant → Get AED 25 Retail voucher.

Ticket → Retail

Buy VIP Admission → Get free souvenir.

Configuration

Source Product/Category

↓

Qualification

↓

Destination Product/Category

↓

Reward

Screen 8 — Reward Selection, Substitution & Customer Choice

Purpose

Control situations where the customer can choose between multiple promotional rewards.

Example

Buy Family Pass and choose ONE:

- Free meal
- Free photo
- Free locker
- Free souvenir

Selection Models

Automatic

TICVAI selects the reward.

Customer Choice

Guest selects from configured options.

Operator Choice

POS/call-center operator selects.

AI Recommended

TICVAI recommends the most relevant reward.

Substitution Rules

If Reward A is unavailable:

A → B → C

Example:

Free T-shirt unavailable

↓

Free cap

↓

AED 30 Retail voucher

Value Protection

Administrators can configure maximum substitute value.

Screen 9 — Advanced Offer Guardrails & Conflict Controls**Purpose**

Prevent advanced offers from generating unintended financial or operational outcomes.

This screen handles **offer-specific safeguards**; the complete cross-promotion stacking hierarchy remains in Board 8.

Controls

- Maximum free items
- Maximum reward value
- Maximum discount
- Maximum applications per basket
- Maximum applications per customer
- Maximum daily redemptions
- Maximum campaign redemptions
- Minimum transaction value
- Minimum margin
- Inventory requirement

Promotion Interaction**Define preliminary behavior:**

- Can combine
- Cannot combine
- Exclusive
- Defer to central stacking engine

Circular Reward Prevention**Prevent configurations such as:**

Product A gives Product B free

Product B gives Product A free

when this could create uncontrolled reward loops.

Approval Trigger**Examples:**

Reward value > AED 500 → Finance approval.

Margin <10% → Commercial Director approval.

Screen 10 — Offer Simulation, Basket Trace & AI Optimization

Purpose

Test complex promotion mechanics before publication.

Test Basket Example

Item	Qty	Unit Price
Adult Admission	2	AED 200
Child Admission	2	AED 150
Souvenir	1	AED 50

Subtotal: AED 750

Applied Promotion

FAMILY4

Buy 4 admission tickets → Cheapest Admission Free.

Calculation

Eligible tickets:

AED 200

AED 200

AED 150

AED 150

Cheapest qualifying ticket:

AED 150

Final

Original Total: **AED 750**

Promotion: **−AED 150**

Final Total: **AED 600**

Rule Trace

Show:

Qualifier

- Admission quantity ≥ 4 ✓
- Eligible venue ✓
- Eligible channel ✓

Reward

- Cheapest admission selected ✓
- Child Ticket AED 150 ✓

Guardrails

- Maximum reward AED 200 ✓
- Margin floor maintained ✓

Simulation Modes

Allow:

- Single transaction
 - Historical transaction replay
 - Sample customer segment
 - Forecast simulation
 - Bulk scenario testing
-

AI Requirements for Board 4**AI shall support the advanced-offer engine by recommending:**

- Optimal Buy X/Get Y combinations
- Appropriate qualifying quantities
- Appropriate reward products
- Complementary products
- Low-cost/high-perceived-value gifts
- Cross-category opportunities
- Reward substitution
- Maximum reward value
- Campaign duration
- Target customer segment

AI should also detect:

- Excessive giveaway cost
- Margin erosion
- Cannibalization
- Poor product pairing
- Inventory risk
- Low-performing rewards
- Unexpected redemption behavior

Example AI Recommendation

Customers purchasing 2 Adult + 1 Child admission have a 38% probability of purchasing F&B. Offering a free child's drink instead of a 10% ticket discount is forecast to improve conversion by 7.2% while reducing promotion cost by AED 18 per transaction.

Key Integrations**Ticketing Catalogue**

Qualifier and reward ticket products.

Promotion Rule Engine — Board 2

Eligibility, thresholds and standard discount calculations.

Coupon Engine — Board 3

Code-triggered advanced offers.

F&B

Menu products, prices and inventory.

Retail

Retail products and stock availability.

Inventory

Free-product availability and substitution.

Membership & Loyalty

Member eligibility and benefits.

CRM

Customer segmentation and personalization.

Pricing

Price basis and margin protection.

Finance

Revenue allocation and promotional cost.

Payment

Final transaction amount.

AI / Analytics

Recommendation and optimization.

Important Architecture Rule

We should maintain a clear distinction:

Board 2 — Promotion Rule Studio

Who/what/when qualifies and what basic discount applies?

↓

Board 3 — Coupons & Codes

Does a code trigger the promotion?

↓

Board 4 — Advanced Mechanics

What product-to-product reward relationship occurs?

↓

Board 8 — Stacking & Conflict Engine

What happens when multiple promotions qualify simultaneously?

This separation prevents duplicated commercial logic.

Matrix Coverage — Board 4

Board 4 directly covers or substantially contributes to:

3.5.4 — re-entry/BOGO and similar promotion mechanics.

3.6.16 — promotions such as Buy X/Get X free and product combination discounts.

3.6.17 — added-value rewards based on spending.

3.6.20-3.6.21 — free ticket/reward entitlements where triggered by qualifying promotional mechanics.

3.6.26 — offers combining tickets with F&B/Retail.

3.6.32 — volume-based dynamic offers including Buy One Get One and 4-for-3.

3.6.34 — cart-level promotion and bundle logic such as Buy 4 Pay 3.

4.1.9 — BOGO based on product type/count.

4.1.10 — the complete Buy X/Get X, Buy X/Get Y, Buy N/Get X, percentage/amount-off and fixed-price mechanics.

4.1.11 — cheapest-item-free logic.

4.1.12 — Retail ↔ F&B cross-category promotional rewards.

Acceptance Criteria

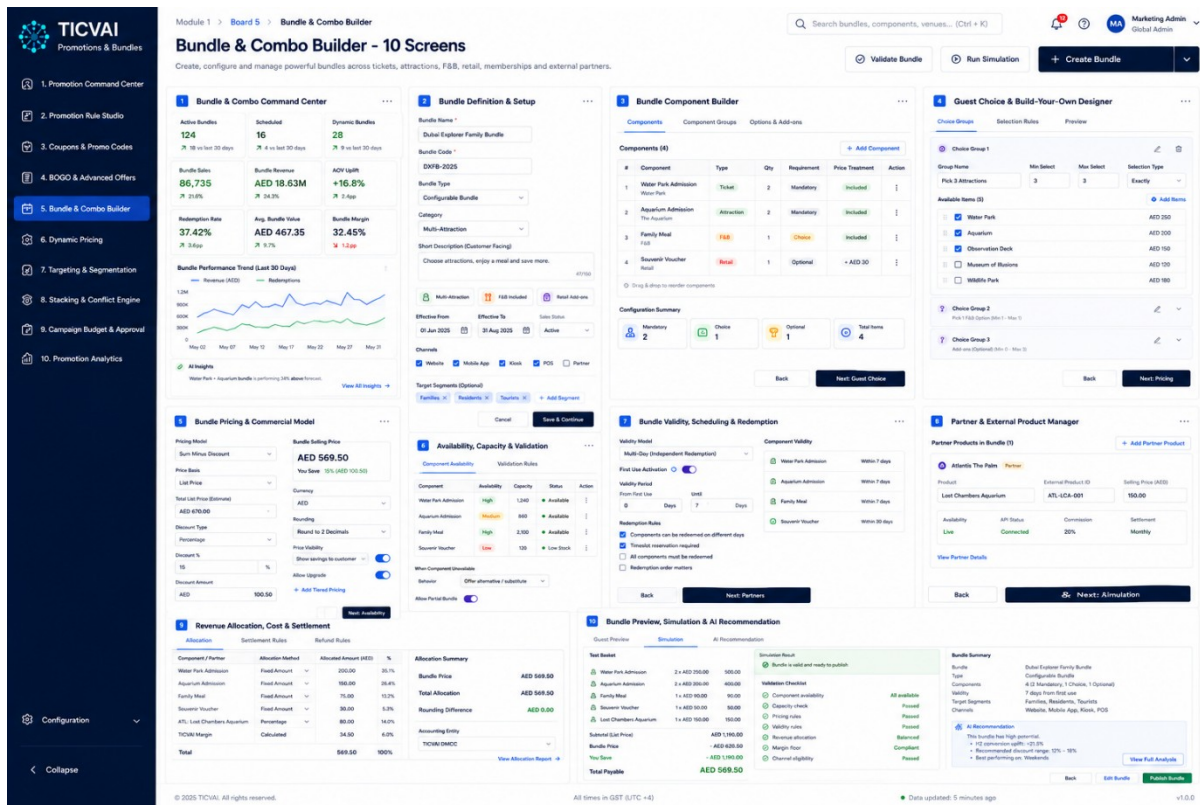
Board 4 shall be considered complete when:

1. Users can configure Buy X Get X.
2. Users can configure Buy X Get Y.
3. Users can configure Buy N Get X.
4. Multiple free/reward quantities are supported.
5. Buy N/Pay X mechanics are supported.
6. Promotion repetition behavior is configurable.
7. Mixed-product quantity qualification is supported.
8. Cheapest-item-free logic is supported.
9. Multiple cheapest-item rewards are supported.
10. Fixed-price/N-for-X promotions are supported.
11. Fixed promotional value can be allocated across underlying products.
12. Spend-and-get-free-product offers are supported.
13. Free tickets, F&B, Retail, experiences and add-ons can be rewards.
14. Inventory availability can affect physical-product rewards.
15. Cross-category Ticketing/F&B/Retail promotions are supported.
16. Customers can choose between configured rewards where allowed.
17. Reward substitution is supported.
18. Maximum reward values and commercial guardrails can be enforced.
19. Circular/uncontrolled reward relationships are prevented.
20. Advanced offers can be simulated before activation.
21. Simulation explains exactly how qualifier and reward calculations occurred.
22. AI can recommend qualifier/reward combinations.
23. AI can estimate revenue, margin and redemption impact.
24. All advanced-offer configurations are versioned and audited.
25. RBAC/PBAC controls govern creation, financial changes, overrides and approval.
26. The engine works consistently across B2C, B2B, POS, Mobile App, Kiosk, Call Center and API channels.
27. Advanced promotion mechanics reuse the central TICVAI promotion architecture rather than duplicate discount, eligibility or stacking logic.

Board 4 is therefore locked as: *BOGO & Advanced Promotion Mechanics* — 10 backend screens.

Module 1 — Promotions & Bundles Management

Board 5 — Bundle & Combo Builder



Board Objective

Board 5 shall provide TICVAI with a centralized **Bundle & Combo Product Configuration Engine** that allows authorized business users to combine multiple products, services, tickets, attractions, F&B items, retail products, memberships, experiences, add-ons, and external partner products into a single sellable commercial offer.

Unlike **Board 4**, which manages conditional mechanics such as *Buy X → Get Y*, Board 5 creates an actual **commercial bundle/combo that the guest purchases as one offer**.

The matrix requires preconfigured and guest-selectable combos, combinations across attractions/complexes, F&B and Retail, independent component rules, configurable/dynamic bundles, external products, and revenue allocation across included components.

Board 5 shall contain **10 backend screens**.

Screen 1 — Bundle & Combo Command Center

Purpose

Provide centralized visibility and management of all bundles and combo products across TICVAI.

KPI Cards

- Active Bundles
- Draft Bundles
- Scheduled Bundles
- Dynamic Bundles

- Fixed Bundles
- Guest-Choice Bundles
- Partner Bundles
- Bundle Sales
- Bundle Revenue
- Average Bundle Value
- Bundle Conversion Rate
- Redemption Rate
- AOV Uplift
- Bundle Margin

Bundle Categories

Filter by:

- Ticket bundle
- Multi-attraction
- Multi-park
- Family package
- Ticket + F&B
- Ticket + Retail
- Ticket + Experience
- Ticket + Parking
- Membership package
- Partner bundle
- Hotel package
- Dynamic bundle
- Build-your-own bundle

Status

- Draft
- Incomplete
- Pending Validation
- Pending Approval
- Scheduled
- Active
- Paused
- Suspended
- Expired
- Archived

AI Insights

Example:

AI Opportunity: Guests purchasing Water Park + Aquarium separately have a 34% overlap. Consider creating a combined offer.

Screen 2 — Bundle Definition & Setup

Purpose

Create the commercial identity and high-level behavior of a bundle.

General Information

Configure:

- Bundle name
- Bundle ID
- Internal description
- Guest-facing description
- Business entity
- Venue
- Bundle category
- Owner
- Campaign
- Effective dates
- Sales status

Bundle Type

Select:

Fixed Bundle

All components predefined.

Water Park + Aquarium + Meal.

Configurable Bundle

Required components plus optional choices.

Water Park + choose one meal.

Build Your Own

Customer selects from permitted categories.

Choose any 3 attractions.

Dynamic Bundle

Components/offers may change according to availability, guest context, or rules.

Partner Bundle

Contains internal and external products.

Selling Behavior

Configure whether the bundle:

- Appears as standalone product.
- Is recommended during checkout.
- Is available only through selected channels.
- Is available only to selected segments.
- Can be purchased independently.
- Requires another product.

Screen 3 — Bundle Component Builder

Purpose

Define exactly what products and services make up the bundle.

Supported Components

- Admission ticket
- Attraction
- Event
- Timeslot
- Experience
- Membership
- Annual pass
- F&B product
- F&B meal package
- Retail product
- Parking
- Locker
- Photo
- Rental
- Voucher
- Gift card
- Service
- External product

Component Table

Component	Qty	Requirement	Price Treatment
Water Park	1	Mandatory	Included
Aquarium	1	Mandatory	Included
Meal	1	Choice	Included
Photo	1	Optional	+AED 30

Component Types

- Mandatory
- Optional
- Choice
- Conditional
- Recommended

Quantity Rules

Configure:

- Fixed quantity

- Minimum
 - Maximum
 - Quantity based on guest count
 - Quantity based on ticket count
-

Screen 4 — Guest Choice & Build-Your-Own Bundle Designer

Purpose

Configure bundles where the guest chooses products from predefined groups.

The matrix specifically requires the guest to be able to choose attractions, experiences, F&B, retail products, or services from predefined categories while maintaining bundle pricing rules.

Example

Dubai Explorer Bundle

Choose 3 of 5 Attractions

- Water Park
- Aquarium
- Observation Deck
- Museum
- Adventure Park

Choose 1 F&B Option

- Meal A
- Meal B
- Meal C

Optional Add-on

- Photo
- Locker
- Parking

Choice Group Configuration

For each group:

- Group name
- Category
- Minimum selections
- Maximum selections
- Required/optional
- Eligible products
- Upgrade products
- Additional charge
- Selection order

Example

Choose exactly **3** attractions.

or

Choose **2-4** experiences.

Screen 5 — Bundle Pricing & Commercial Model

Purpose

Determine how the bundle is commercially priced.

Pricing Models

Fixed Bundle Price

Bundle = AED 499.

Sum Minus Discount

Components total AED 600

Bundle discount 15%

Bundle = AED 510.

Component Pricing

Each component retains configured price.

Starting From

From AED 399 depending on guest selections.

Tiered Bundle Price

2 attractions = AED 299

3 attractions = AED 399

4 attractions = AED 469.

Dynamic Bundle Price

Calculated through TICVAI's pricing engine.

Pricing Configuration

- Base price
- Currency
- Discount %
- Discount amount
- Minimum price
- Maximum price
- Price floor
- Margin floor
- Guest-specific price
- Channel-specific price

Upgrade Charges

Example:

Standard meal included.

Guest chooses Premium Meal:

+AED 30

Screen 6 — Bundle Availability, Capacity & Validation

Purpose

Ensure that TICVAI does not sell a bundle unless all required components can actually be fulfilled.

This is a direct requirement of the matrix: each bundle component maintains independent inventory, capacity, validity, redemption rules, and availability, and the system validates included products before confirming the sale.

Component Availability

For each component display:

- Available
- Low availability
- Sold out
- Suspended
- Unpublished
- Invalid date
- Capacity unavailable

Validation Sequence

Before confirmation:

Bundle selected

↓

Check mandatory components

↓

Check inventory

↓

Check capacity

↓

Check schedule/timeslot

↓

Check validity

↓

Check pricing

↓

Confirm Bundle

Failure Behavior

If one component is unavailable:

- Prevent sale
- Hide bundle
- Offer substitute
- Allow alternate date
- Allow alternate timeslot
- Remove optional component

- Recommend another bundle

Capacity Rules

Bundle sale must consume capacity from the **underlying component**, not create false independent capacity.

Screen 7 — Bundle Validity, Scheduling & Redemption Rules

Purpose

Control when bundle components may be consumed and whether they must be redeemed together or separately.

Validity Models

Same-Day

All components used on one day.

Multi-Day

Example:

Use any included attraction within 7 days.

Relative

Bundle valid 30 days from first activation.

Fixed Period

Valid 1 June–31 August.

Redemption Models

- All components together
- Independent redemption
- Sequential redemption
- First-use activation
- Scheduled redemption
- Timeslot reservation required

Example

3 Attraction Pass

First attraction:

1 June

Remaining attractions valid until:

7 June

Component Rules

Each component may maintain:

- Own validity
- Own capacity
- Own timeslot
- Own redemption count
- Own entitlement
- Own access rule

This is particularly important for multi-park and multi-attraction products.

Screen 8 — Partner & External Product Bundle Manager

Purpose

Allow TICVAI bundles to include products or services owned by external operators.

The matrix requires combinations with external products/services and bundles across different destinations, including systems that may use different databases.

External Partners

Examples:

- Hotel
- OTA
- Airline
- Tour operator
- Transportation provider
- Restaurant
- External attraction
- Tourism destination

Partner Product Configuration

- Partner
- External product ID
- Product name
- API source
- Availability source
- External price
- TICVAI selling price
- Commission
- Settlement rule
- Cancellation rule
- Redemption method

Integration Status

Show:

- Connected
- Available
- Degraded
- API Error
- Product unavailable
- Mapping error

Example

Dubai Weekend Package

- TICVAI Water Park

- External Hotel Night
- External Desert Safari
- TICVAI Meal Voucher

sold as one commercial package.

Screen 9 — Revenue Allocation, Cost & Settlement Rules

Purpose

Determine how bundle revenue is allocated across its components.

This is explicitly required by the matrix for allocation across products, attractions, departments, partners, operators, and accounting entities.

Example

Customer pays:

AED 500

Allocation:

Component	Allocation
Water Park	AED 200
Aquarium	AED 150
F&B	AED 75
External Partner	AED 50
TICVAI/Operator Margin	AED 25

Allocation Methods

- Fixed amount
- Percentage
- Proportional list price
- Weighted allocation
- Cost-plus
- Contractual partner allocation
- Redemption-based allocation

Financial Configuration

- Revenue account
- Department
- Cost center
- Legal entity
- Tax treatment
- Partner payable
- Commission

- Settlement cycle

Refund Allocation

The same allocation structure must support:

- Full refund
- Partial refund
- Component refund
- Unredeemed component refund

Actual accounting execution should be handled through the Finance integration.

Screen 10 — Bundle Preview, Simulation & AI Recommendation

Purpose

Provide end-to-end validation before a bundle is published.

Bundle Preview

Administrators should preview the guest journey for:

- B2C
- Mobile App
- POS
- Kiosk
- B2B
- Partner channel

Simulation Example

Family Adventure Bundle

2 Adult Tickets
2 Child Tickets
1 Family Meal
1 Parking

Individual value:

AED 720

Bundle price:

AED 599

Guest saving:

AED 121 / 16.8%

Estimated margin:

31.4%

Validation

Check:

- Component availability ✓
- Capacity ✓
- Pricing ✓
- Validity ✓

- Revenue allocation ✓
- Channel assignment ✓
- Tax ✓
- Partner connection ✓
- Margin floor ✓

Publication Result

READY TO PUBLISH

or show blocking issues.

AI Requirements for Board 5

AI shall analyze historical and contextual data to recommend commercially attractive bundle combinations.

The matrix explicitly requires bundle recommendations based on historical sales, guest behavior, demographics, seasonality, attraction popularity, and purchasing trends.

AI shall recommend:

- Products commonly purchased together.
- Attractions likely to convert as bundles.
- F&B additions.
- Retail additions.
- Experiences.
- Membership upgrades.
- Optimal number of components.
- Bundle price.
- Discount level.
- Target customer segment.
- Channel.
- Season.
- Validity period.

Example

AI Bundle Opportunity

42% of Water Park visitors purchase Aquarium tickets within seven days, while 31% purchase an F&B meal.

Recommended bundle:

Water Park + Aquarium + Meal

Suggested price: **AED 399**

Predicted conversion uplift: **+14.8%**

Predicted AOV uplift: **+21.3%**

AI Guardrail

AI may create a **recommended draft**, but should not autonomously publish the bundle without configured approval.

Key Integrations

Ticketing Catalogue

Ticket and admission products.

Event / Resource / Seat Management

Scheduled products and capacity.

F&B

Menu products, pricing and fulfillment.

Retail

Retail products and inventory.

Inventory

Physical-product availability.

Membership & Loyalty

Membership components and eligibility.

Pricing / Dynamic Pricing

Bundle and component pricing.

Promotion Engine

Additional promotion eligibility.

Access Control

Redemption of bundle entitlements.

B2B / Reseller

Partner availability and distribution.

External API Layer

External products and availability.

Finance

Revenue allocation and settlement.

Payment

Bundle transaction payment.

Reporting / AI

Bundle performance and recommendations.

Important Architecture Principle

There must be a strict distinction between **Bundle Product** and **Promotion**.

Bundle

“Water Park + Aquarium + Meal = AED 399”

This is a **sellable commercial product composed of multiple components**.

Promotion

“Buy Water Park → Get Meal Free”

This is a **conditional promotional relationship** handled by Board 4.

Therefore:**Board 4 — Advanced Promotions**

Qualification → Reward

versus

Board 5 — Bundle Builder

Components → Package → Price → Sell

This distinction will prevent major duplication later.

Matrix Coverage — Board 5

Board 5 directly covers or substantially contributes to:

3.5.8 — independent inventory, capacity, validity, redemption and availability for bundle components.

3.5.9 — presenting bundles according to configurable conditions.

3.5.10 — dynamic/customer-selectable bundles across attractions, experiences, F&B, Retail and services.

3.5.11 — revenue allocation across products, attractions, departments, partners and accounting entities.

3.5.12 — foundation for bundle performance analytics.

3.5.13 — AI bundle recommendations.

3.6.3-3.6.4 — combos across attractions, Retail and F&B, including predefined and guest-choice combinations.

3.6.26 — ticket packages with F&B/Retail.

3.7.1 — externally managed bundle products.

3.7.3-3.7.4 — combinations with external products/destinations and revenue splitting.

4.1.3-4.1.5 — bundle-related cross-sell/commercial product relationships where the outcome is an actual package.

Acceptance Criteria

Board 5 shall be complete when:

1. Fixed bundles can be created.
2. Configurable bundles can be created.
3. Build-your-own bundles can be created.
4. Dynamic bundles are supported.
5. Partner/external bundles are supported.
6. Ticketing, F&B, Retail, membership, experience, parking and service products can coexist within one bundle.
7. Components can be mandatory, optional, selectable or conditional.
8. Choice groups support minimum and maximum selections.
9. Bundle pricing can be fixed, calculated, tiered or dynamic.
10. Component upgrade charges are supported.
11. Each component maintains its own availability and capacity.
12. Required components are validated before sale.
13. Sold-out components can block, substitute or modify the bundle according to configuration.
14. Components can maintain independent validity and redemption rules.
15. Multi-day and first-use bundle validity is supported.
16. External products can be incorporated through APIs.
17. External availability and product mapping can be monitored.
18. Bundle revenue can be allocated across products, departments, entities and partners.
19. Revenue allocation supports settlement and refund scenarios.

20. Users can preview the bundle across supported sales channels.
21. Bundles can be simulated before publication.
22. TICVAI identifies blocking configuration problems before publication.
23. AI recommends bundle combinations based on behavioral and commercial data.
24. AI recommends pricing, target segments and timing.
25. AI recommendations remain governed by approval policies.
26. Bundle configuration is versioned and audited.
27. RBAC/PBAC governs bundle creation, pricing, revenue allocation and publication.
28. Bundle components remain linked to their source product definitions rather than unnecessarily duplicating SKUs.
29. The engine supports multi-tenant, multi-venue, multi-currency and multi-entity operation.
30. The architecture supports new product/component types without redesigning the Bundle Engine.

Board 5 is locked as: *Bundle & Combo Builder* — 10 backend screens.

Module 1 — Promotions & Bundles Management

Board 6 — Dynamic Bundle & Inventory Engine



Board Objective

Board 6 shall provide TICVAI with the **runtime intelligence and operational engine behind bundles**.

Board 5 answers:

“What products and choices make up this bundle?”

Board 6 answers:

“Can TICVAI sell this bundle right now, what components should be offered, what should happen when availability changes, and how should the bundle dynamically adapt?”

This distinction is important. The matrix specifically requires each bundle component to maintain independent **inventory, capacity, validity, redemption rules, and availability**, with all required components validated before confirmation. It also requires dynamic bundles where guests select products from predefined categories and bundle offers that respond to configurable conditions.

Board 6 shall contain **10 backend screens**.

Screen 1 — Dynamic Bundle Operations Command Center

Purpose

Provide real-time visibility into the operational health of all active bundles.

KPI Cards

- Active Dynamic Bundles

- Sellable Bundles
- Partially Available Bundles
- Unavailable Bundles
- Bundles with Low Capacity
- Components Sold Out
- Substitutions Triggered
- Bundle Sales Today
- Failed Bundle Attempts
- Capacity Reserved
- Revenue at Risk
- Recovered Revenue

Bundle Health

Each bundle receives:

- Healthy
- Warning
- Critical
- Unavailable

Example

Family Adventure Bundle

Water Park — Available

Aquarium — Available

Family Meal — Low Stock

Photo — Available

Bundle Health: WARNING

Operational Alerts

Examples:

Aquarium capacity has fallen below 10%.

Family Meal is unavailable at Venue A.

Alternative Meal B automatically activated.

Partner component API has been unavailable for 5 minutes.

Screen 2 — Component Inventory & Availability Matrix

Purpose

Provide one centralized matrix showing availability for every component within every active bundle.

Matrix Example

Component	Inventory	Capacity	Schedule	Status
Water Park	Unlimited	1,240	Open	Available

Component	Inventory	Capacity	Schedule	Status
Aquarium	Unlimited	420	Open	Available
Family Meal	38	—	Open	Low
Souvenir	0	—	Open	Sold Out
Partner Safari	72	72	Open	Available

Availability Sources

TICVAI shall retrieve availability from:

- Ticket inventory
- Attraction capacity
- Event capacity
- Seat inventory
- Timeslots
- F&B availability
- Retail stock
- Resource availability
- Parking
- Rental
- External partner API

Component Status

- Available
- Limited
- Low
- Sold Out
- Closed
- Suspended
- Unpublished
- API Unavailable

Screen 3 — Bundle Sellability & Dependency Rule Engine

Purpose

Determine whether the overall bundle can be sold based on the state of its underlying components.

Rule Example

IF

Water Park = Available

AND

Aquarium = Available

AND

At least 1 Meal Option = Available

THEN

Bundle = Sellable

Dependency Types

Support:

- All components required
- At least X of Y
- At least one from category
- Optional component
- Conditional component
- Substitute allowed
- Partner component required

Example

Bundle contains:

- Water Park — Mandatory
- Aquarium — Mandatory
- Meal — Choose 1 of 3
- Photo — Optional

The bundle remains sellable if the Photo is unavailable.

It becomes unavailable if Water Park is unavailable.

Rule Visualization

Use:

Component → Dependency → Bundle Status

to make the logic understandable to administrators.

Screen 4 — Capacity Pool & Reservation Manager

Purpose

Manage how bundle sales consume capacity from underlying products.

This is particularly important because a bundle **must not create artificial inventory** separate from the actual attraction/product capacity.

Example

Aquarium remaining capacity:

100

Standalone sales:

60

Bundle reservations:

25

Held:**5****Remaining sellable:****10****Capacity Sources**

- Shared pool
- Dedicated bundle allocation
- Channel allocation
- Partner allocation
- Event capacity
- Timeslot capacity
- Seat inventory
- Resource capacity

Reservation Controls**Configure:**

- Temporary reservation
- Hold duration
- Release timeout
- Hard allocation
- Soft allocation
- Overbooking policy
- Waitlist behavior

Example**Customer starts checkout:****Bundle components temporarily held for:****10 minutes****If payment fails or checkout expires:****Capacity automatically released.**

Screen 5 — Dynamic Component Substitution Engine

Purpose

Automatically replace unavailable bundle components according to predefined commercial rules.

Example**Original:****Family Meal A**

Unavailable

↓

Substitution Priority:

1. Family Meal B

2. Family Meal C
3. AED 50 F&B Voucher
4. Remove component and adjust price

Configuration

For each component define:

- Primary component
- Alternative 1
- Alternative 2
- Alternative 3
- Fallback action

Substitution Conditions

Substitution may occur when:

- Sold out
- Capacity exhausted
- Product suspended
- Venue closed
- Schedule unavailable
- External API unavailable
- Inventory below threshold

Commercial Treatment

Define whether substitute:

- Maintains same bundle price
 - Adds surcharge
 - Reduces bundle price
 - Requires customer approval
 - Requires operator approval
-

Screen 6 — Dynamic Bundle Rule & Composition Engine

Purpose

Allow the actual composition of a bundle to change dynamically according to business and guest conditions.

This builds upon the matrix requirement for dynamic bundles where guests select attractions, experiences, F&B, Retail, or services from predefined categories.

Conditions

Bundle composition may depend on:

- Guest segment
- Membership
- Loyalty tier
- Purchase history
- Number of guests

- Guest type
- Sales channel
- Venue
- Season
- Day
- Time
- Capacity
- Inventory
- Campaign
- Product popularity

Example**Family Segment****Offer:**

Water Park + Aquarium + Family Meal

Couple Segment**Offer:**

Water Park + Premium Dinner + Photo

Annual Pass Holder**Offer:**

Experience + F&B + Retail rather than another admission ticket.

No-Code Rule**IF**

Segment = Family

AND

Guests ≥ 4

AND

Aquarium Capacity >20%

THEN

Include Aquarium as recommended component.

Screen 7 — Real-Time Availability & Checkout Validation

Purpose

Perform the final authoritative validation immediately before transaction confirmation.

This is essential because availability may change between browsing and payment.

Validation Sequence**Guest selects bundle**

↓

Initial availability check

↓

Reserve components

↓
 Guest completes selections
 ↓
 Price calculated
 ↓
 Payment initiated
 ↓
FINAL AVAILABILITY CHECK
 ↓
 Payment authorization/capture
 ↓
 Confirm bundle
 ↓
 Commit inventory/capacity

Validation Checklist

- Product active
- Inventory available
- Capacity available
- Timeslot available
- Resource available
- Price valid
- Promotion valid
- Partner component valid
- Component mapping valid

Failure Handling

If validation fails:

- Recheck availability
- Offer substitute
- Offer alternative time
- Offer alternative date
- Remove optional component
- Recommend another bundle
- Cancel checkout safely

Screen 8 — Bundle Availability by Channel, Venue & Partner

Purpose

Control where a bundle is sellable based on operational availability.

Channels

- B2C
- B2B

- POS
- Mobile POS
- Mobile App
- Kiosk
- Call Center
- API
- OTA
- Reseller
- Partner

Availability Matrix

Bundle	B2 C	POS	B2 B	OTA	Partner
Family Pack	Live	Live	Live	Off	Live
Explorer	Live	Live	Live	Live	Live
VIP Combo	Live	Live	Off	Off	Off

Venue Rules

Configure:

- Venue-specific availability
- Attraction-specific availability
- Operating area
- Country/market
- Sales location

Partner Controls

Partners may receive:

- Dedicated allocation
- Shared allocation
- Capacity ceiling
- Booking cutoff
- Release period

Screen 9 — Bundle Availability Forecast, Alerts & Recovery

Purpose

Predict bundle availability problems before they affect sales.

Forecasting

Analyze:

- Historical demand

- Current booking velocity
- Inventory
- Capacity
- Timeslot utilization
- Seasonality
- Day of week
- Campaign activity
- Partner reservations

Forecast Examples

Aquarium capacity is forecast to reach 100% at 16:30.

Family Meal inventory is likely to run out within 2.5 hours.

Partner Safari inventory has a 78% probability of exhaustion by tomorrow.

Alert Levels

- Information
- Warning
- Critical

Automated Recovery Recommendations

TICVAI may recommend:

- Activate substitute
- Remove component
- Switch choice group
- Restrict channel
- Reduce bundle allocation
- Increase alternative inventory
- Recommend another timeslot
- Pause bundle

Revenue at Risk

Show:

Potential Bundle Revenue at Risk

AED 186,500

Potential Revenue Recoverable Through Substitution

AED 142,000

Screen 10 — Dynamic Bundle Simulation & AI Optimization

Purpose

Test how a bundle behaves under different operational scenarios before activating dynamic rules.

Scenario Simulation

Administrators can simulate:

Scenario A

All components available.

Scenario B

Aquarium sold out.

Scenario C

Meal inventory low.

Scenario D

Partner API unavailable.

Scenario E

Attraction capacity at 95%.

Example Simulation

Original:

Water Park

Aquarium

Meal A

Photo

Aquarium unavailable.

TICVAI Response

Rule evaluation:

Aquarium required → Substitute allowed.

↓

Alternative:

Observation Deck available.

↓

Price difference:

+AED 20

↓

Customer approval required.

↓

Bundle remains sellable.

Simulation Output

Show:

- Bundle status
- Components selected
- Substitutions
- Price impact
- Margin impact
- Capacity impact
- Customer impact
- Revenue impact

AI Requirements for Board 6

AI should provide the intelligence layer for dynamic bundle operations.

AI shall forecast:

- Component demand
- Inventory exhaustion
- Capacity exhaustion
- Timeslot demand
- Bundle demand
- Partner inventory risk

AI shall recommend:

- Component substitutions
- Alternative attractions
- Alternative F&B
- Alternative Retail
- Allocation changes
- Capacity changes
- Channel restrictions
- Bundle composition changes
- Availability windows
- Alternative bundles

AI Optimization Example

AI Availability Risk

Aquarium is forecast to sell out between 16:00–18:00.

1,240 upcoming Explorer Bundle transactions could be affected.

Recommended action:

Automatically replace Aquarium with Observation Deck for new purchases between 16:00–18:00.

Expected protected revenue: **AED 184,000**

AI Commercial Consideration

AI should consider both:

Operational feasibility

and

Commercial impact

before recommending substitutions.

For example, it should not recommend replacing a premium AED 200 experience with a low-value AED 30 item simply because it is available.

Automation & Human Control

Board 6 should support three automation levels.

Level 1 — Recommend Only

AI identifies the problem and recommends action.

Human approves.

Level 2 — Pre-Approved Automation**Example:**

If Meal A unavailable → automatically switch to Meal B.

No approval required because the rule was previously approved.

Level 3 — Emergency Protection**Example:**

If mandatory component unavailable → automatically stop new bundle sales.

This prevents TICVAI from knowingly selling an impossible bundle.

All automated actions must be audited.

Key Integrations**Ticketing Catalogue**

Product status and inventory.

Capacity Management

Attraction/event capacity.

Seat Management

Seat availability where included.

Event Management

Schedule and performance availability.

Resource Management

Resource availability.

F&B

Menu availability.

Retail / Inventory

Physical inventory.

Membership

Membership-based components.

Access Control

Entitlement/redemption state.

Pricing / Dynamic Pricing

Real-time pricing.

Promotions

Promotion qualification.

Payment

Checkout/payment coordination.

B2B / Reseller

Partner allocation.

External API Layer

External availability.

Finance

Commercial impact.

AI / Analytics

Forecasting and optimization.

Important Architecture Principle

Boards 5 and 6 must remain separate.

Board 5 — Bundle & Combo Builder

Defines:

WHAT IS THE BUNDLE?

Components
Choices
Price
Validity
Revenue allocation
Partner products

Board 6 — Dynamic Bundle & Inventory Engine

Determines:

CAN WE SELL IT RIGHT NOW, AND WHAT SHOULD HAPPEN IF CONDITIONS CHANGE?

Availability
Capacity
Inventory
Dependencies
Substitution
Runtime validation
Forecasting
Dynamic composition

This separation will make the TICVAI architecture much cleaner and more scalable.

Matrix Coverage — Board 6

Board 6 directly or substantially covers:

3.5.8 — independent inventory, capacity, validity, redemption rules and availability for every bundle component, plus pre-sale validation.

3.5.9 — bundle presentation according to conditions such as ticket quantity, guest type, loyalty, membership, channel, season, location and campaign.

3.5.10 — dynamic bundle composition using predefined categories.

3.5.13 — AI recommendations using historical sales, guest behavior, seasonality, popularity and purchasing trends.

3.6.3-3.6.4 — operational support for preconfigured and guest-selectable combos.

3.6.26 — packages involving Ticketing, F&B and Retail.

3.7.1 — externally managed bundle components.

3.7.3-3.7.4 — external product/destination combinations.

4.1.2-4.1.5 — operational foundation for dynamically presenting relevant upgrade, cross-sell and bundle opportunities.

Some of these requirements span multiple boards intentionally: **Board 5 defines the commercial bundle; Board 6 executes and protects it operationally.**

Acceptance Criteria

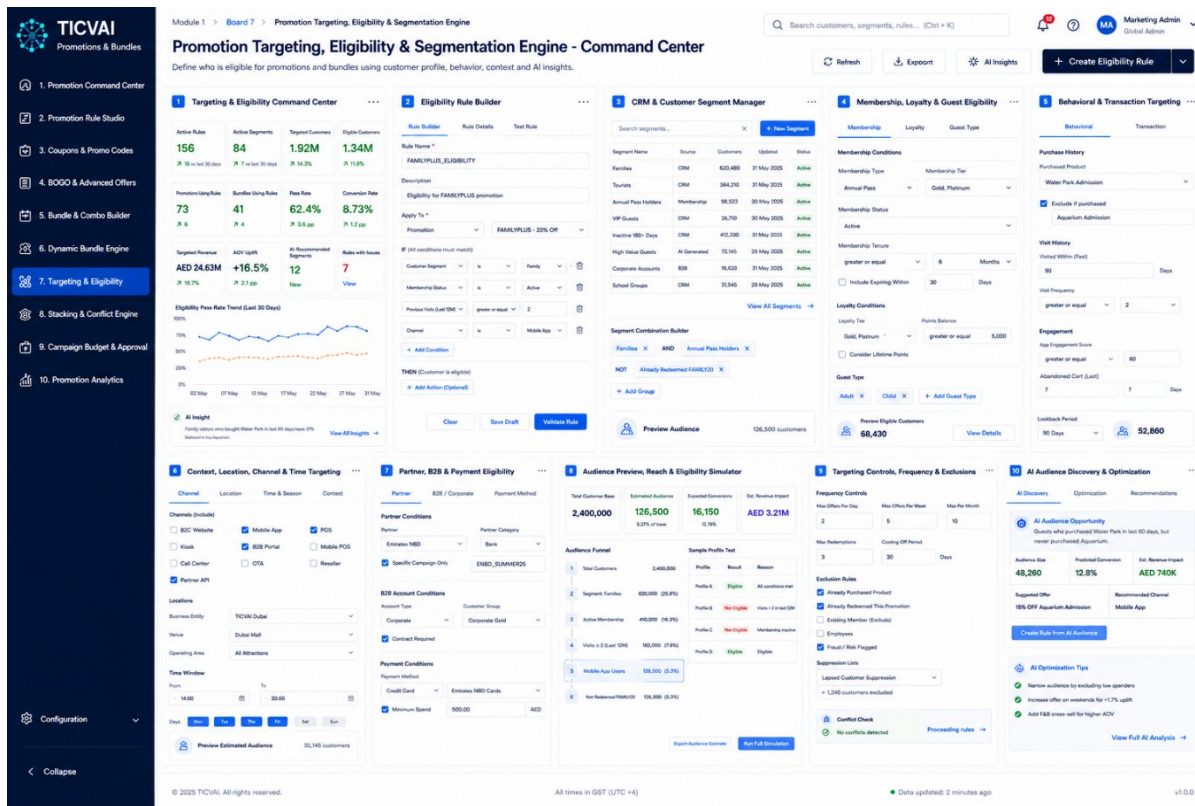
Board 6 shall be considered complete when:

1. TICVAI can determine real-time sellability of every bundle.
2. Every component retains its independent inventory/capacity source.
3. Bundle sales consume actual underlying capacity.
4. Mandatory and optional components are treated differently.
5. Complex component dependencies are supported.
6. Choice-group availability is validated.
7. Temporary inventory/capacity holds are supported during checkout.
8. Expired checkout holds are automatically released.
9. Component substitutions can be configured.
10. Multiple substitution priorities are supported.
11. Price differences caused by substitutions are handled correctly.
12. Customer/operator approval can be required for substitutions.
13. Bundle composition can dynamically respond to guest and operational conditions.
14. Final availability is revalidated before transaction confirmation.
15. Failed final validation cannot create an invalid booking.
16. Alternative dates, times, components, or bundles can be proposed.
17. Availability can differ by channel, venue, and partner.
18. Partner allocations and booking cutoffs are supported.
19. TICVAI forecasts inventory and capacity exhaustion.
20. Revenue-at-risk calculations are available.
21. AI recommends recovery actions.
22. Operational scenarios can be simulated before activating rules.
23. AI recommendations consider both availability and commercial value.
24. Pre-approved automation can execute configured substitutions.
25. Mandatory-component failure can automatically stop sales where configured.
26. Every automatic and manual action is audited.
27. RBAC/PBAC governs overrides and automation policies.
28. Runtime rules operate consistently across all supported sales channels.
29. The architecture supports internal and external component sources.
30. New inventory/capacity sources can be introduced without redesigning the bundle engine.

Board 6 is locked as: *Dynamic Bundle & Inventory Engine* — 10 backend screens.

Module 1 — Promotions & Bundles Management

Board 7 — Promotion Targeting, Eligibility & Segmentation Engine



Board Objective

Board 7 shall provide TICVAI with the centralized **Promotion Targeting & Eligibility Engine** that determines exactly **who can receive, see, purchase, or redeem a promotion or bundle, under which commercial and contextual conditions.**

This board is different from Board 2.

Board 2 — Promotion Rule Studio defines:

What commercial benefit applies?

Board 7 — Targeting & Eligibility Engine defines:

Who should receive or qualify for that benefit?

The matrix requires promotions based on customer profile, purchase history, demographics, loyalty/membership, guest type, transaction behavior, location, channel, season, partner, campaign, payment method, and CRM segments. It also requires personalized and AI-driven promotion recommendations.

Board 7 shall contain **10 backend screens.**

Screen 1 — Targeting & Eligibility Command Center

Purpose

Provide centralized visibility into all promotion audiences, eligibility rules, segments, targeting strategies, and their performance.

KPI Cards

- Active Targeting Rules
- Active Segments
- Promotions Using Targeting
- Bundles Using Targeting
- Eligible Customers
- Targeted Customers
- Personalized Offers
- Eligibility Pass Rate
- Conversion Rate
- Targeted Revenue
- AOV Uplift
- AI-Recommended Segments

Targeting Categories

Filter by:

- Guest type
- CRM segment
- Membership
- Loyalty
- Demographic
- Behavioral
- Transaction
- Geographic
- Channel
- Partner
- B2B
- Payment
- Contextual
- AI-generated

Rule Health

Each rule shows:

- Healthy
- Warning
- Conflict
- No Audience
- Oversized Audience

- Expired
- Missing Data

AI Insights

Example:

Audience Opportunity: Families who visited Water Park during the previous 90 days have a 31% likelihood of purchasing Aquarium admission.

Screen 2 — Eligibility Rule Builder

Purpose

Provide a no-code rule engine for determining promotion eligibility.

Rule Structure

Use:

IF → AND/OR → THEN ELIGIBLE

or

IF → THEN EXCLUDE

Example

IF Customer Segment = Family
 AND Membership Status = Active
 AND Previous Visits ≥ 2
 AND Channel = Mobile App
 THEN Eligible for FAMILYPLUS.

Available Conditions

- Guest type
- Customer segment
- Age/category
- Membership
- Loyalty tier
- Purchase history
- Visit history
- Transaction value
- Product purchased
- Channel
- Venue
- Location
- Partner
- Date/time
- Payment method
- Campaign
- Customer/account attributes

Logic

Support:

- AND
 - OR
 - NOT
 - Nested groups
 - Include
 - Exclude
 - Multiple condition sets
-

Screen 3 — CRM & Customer Segment Manager

Purpose

Connect promotion eligibility directly to TICVAI CRM segmentation.

The board should **consume CRM segments rather than recreate CRM functionality**.

Segment Sources

- TICVAI CRM
- Imported segment
- External CRM/CDP
- Membership
- Loyalty
- B2B accounts
- Marketing automation

Example Segments

- Families
- Tourists
- UAE Residents
- Annual Pass Holders
- VIP Guests
- Lapsed Visitors
- Frequent Visitors
- High-Value Guests
- First-Time Guests
- School Groups
- Corporate Customers

Segment Information

Show:

- Segment name
- Source
- Estimated audience
- Last refreshed
- Promotions using segment
- Conversion

- Revenue
- Status

Segment Combination

Allow:

Families

AND

Annual Pass Holders

BUT NOT

Customers who already redeemed FAMILY20

Screen 4 — Membership, Loyalty & Guest Eligibility

Purpose

Configure targeting based on membership, loyalty status, guest categories, and entitlement relationships.

Membership Conditions

- Membership type
- Membership tier
- Membership status
- Membership start date
- Renewal status
- Expiry date
- Membership tenure

Loyalty Conditions

- Loyalty tier
- Points balance
- Points earned
- Points redeemed
- Loyalty activity
- Tier progression

Guest Conditions

- Adult
- Child
- Senior
- Family
- Student
- Resident
- Tourist
- Group
- VIP

Example

Gold Members receive 20% off selected experiences.

Tiered Benefit Example

Tier	Benefit
Bronze	5%
Silver	10%
Gold	15%
Platinum	20%

The commercial benefit remains configured through Board 2; Board 7 provides the eligibility.

Screen 5 — Behavioral & Transaction Targeting

Purpose

Target promotions according to what the guest has previously purchased or done.

Behavioral Conditions

- Previous product purchased
- Previous attraction visited
- Visit frequency
- Days since last visit
- Previous promotion redemption
- Abandoned cart
- Booking frequency
- Purchase frequency
- Average transaction value
- Total customer value
- Product affinity

Examples

Purchased Water Park but never Aquarium.

Visited ≥ 3 times in previous 12 months.

No visit within previous 180 days.

Basket abandoned within previous 24 hours.

Spent more than AED 2,000 in previous year.

Transaction Conditions

- Lifetime spend
- Average basket
- Number of transactions
- Last purchase
- Purchase channel

- Product mix

Lookback Period

Configure:

- Last 7 days
 - 30 days
 - 90 days
 - 12 months
 - Lifetime
 - Custom period
-

Screen 6 — Context, Location, Channel & Time Targeting

Purpose

Determine promotion eligibility according to the customer's current commercial context.

Channel Conditions

- B2C Website
- Mobile App
- POS
- Mobile POS
- Kiosk
- B2B
- Call Center
- API
- OTA
- Reseller
- Partner

Location Conditions

- Business entity
- Venue
- Attraction
- Operating area
- Market/country
- Sales location

Temporal Conditions

- Purchase date
- Visit date
- Day
- Time
- Season
- Event
- Holiday

- Campaign period

Example

Mobile App customers at Venue A between 14:00–17:00 receive a targeted F&B offer.

Contextual Conditions

Support:

- Current basket
 - Product currently viewed
 - Current booking
 - Visit state
 - In-venue state where available
-

Screen 7 — Partner, B2B & Payment Eligibility

Purpose

Configure eligibility for partner, corporate, reseller, B2B, and payment-related campaigns.

Partner Conditions

- Partner
- Partner category
- Corporate account
- Employer
- Hotel
- Travel agency
- Tour operator
- School
- Government entity
- Bank

B2B Conditions

- B2B account
- Master account
- Sub-account
- Contract
- Customer group
- Market
- Sales channel

Payment Eligibility

Promotions may depend on:

- Payment method
- Eligible card program
- Bank
- Wallet
- Loyalty payment

- Gift card
- Approved payment partner

Example

Bank ABC cardholder + B2C channel + AED 500 minimum spend → eligible.

Sensitive card/payment validation remains within **Payment Orchestration**. Board 7 should receive only the eligibility result or permitted payment metadata.

Screen 8 — Audience Preview, Reach & Eligibility Simulator

Purpose

Allow administrators to understand exactly who will qualify **before activating the targeting rule**.

This is a critical safeguard.

Audience Preview**Show:****Total customer population**

2,400,000

↓

Family segment

620,000

↓

Visited within 12 months

310,000

↓

No Aquarium purchase

182,000

↓

Mobile App enabled

126,500

Final Audience

126,500 eligible customers

Audience Metrics

- Estimated audience
- Percentage of customer base
- Historical conversion
- Historical AOV
- Expected redemptions
- Estimated promotion cost
- Estimated revenue

Sample Profiles

Authorized users may test representative/sample profiles:

Profile A → Eligible

Profile B → Not Eligible

Eligibility Explanation

Example:

Customer failed because Membership Status = Expired.

This explanation capability will be very useful for customer service and operational troubleshooting.

Screen 9 — Targeting Conflict, Frequency & Exclusion Controls

Purpose

Prevent customers from being over-targeted and prevent inappropriate promotional eligibility.

Frequency Controls

Configure:

- Maximum offers per day
- Maximum offers per week
- Maximum campaigns per month
- Maximum redemptions
- Cooling-off period
- Repeat campaign restriction

Exclusions

Support:

- Already purchased product
- Already redeemed promotion
- Existing member
- Employee
- Specific CRM segment
- Fraud/risk status
- Account type
- Partner restriction
- Product ownership

Suppression Lists

Administrators may maintain governed suppression groups such as:

- Campaign exclusions
- Partner exclusions
- Operational exclusions

Conflict Example

Customer qualifies for:

Family Campaign

and

VIP Campaign

Board 7 determines whether they are eligible for each.

Board 8 subsequently determines which promotion actually wins or whether they can stack. That separation is important.

Screen 10 — AI Audience Discovery & Targeting Optimization

Purpose

Use TICVAI's AI layer to identify customer audiences and promotion opportunities that business users may not have manually defined.

AI Audience Discovery

AI analyzes permitted data such as:

- Historical transactions
- Product affinity
- Visit frequency
- Basket composition
- Promotion response
- Membership
- Loyalty
- Channel behavior
- Seasonality
- Customer segment
- Campaign history

AI Recommendations

Example:

New Audience Opportunity

Guests who purchased Water Park admission during the previous 60 days but have never purchased Aquarium admission.

Audience: **48,260**

Suggested offer: **15% Aquarium discount**

Predicted conversion: **12.8%**

Estimated incremental revenue: **AED 740K**

Similar Audience

AI may identify:

Customers behaving similarly to guests who responded strongly to Campaign A.

AI Optimization

Recommend:

- Narrow audience
- Expand audience
- Exclude low-value segment

- Change eligibility
 - Change channel
 - Change timing
 - Change promotion
 - Create new segment
 - Reduce frequency
-

AI Requirements for Board 7

AI should be deeply integrated into this board because targeting is one of the strongest areas for TICVAI's AI-first positioning.

AI shall support:

- Audience discovery
- Customer affinity scoring
- Purchase propensity
- Promotion-response probability
- Cross-sell propensity
- Upsell propensity
- Membership-upgrade propensity
- Churn/lapse targeting where permitted
- Segment optimization
- Audience-size prediction
- Conversion prediction
- Revenue prediction
- Promotion-cost prediction

Explainable AI

Every AI recommendation should explain:

Why this audience?

For example:

This segment was recommended because customers have:

- Visited Water Park ≥ 2 times
- Purchased F&B in 71% of visits
- Never purchased Aquarium
- High Mobile App engagement

Business users should not simply receive a black-box audience.

Privacy & Governance

This board handles customer targeting, so privacy controls are particularly important.

Principles

TICVAI should:

- Use only permitted customer attributes.
- Respect customer consent/preferences where applicable.
- Respect configured marketing suppression.

- Apply tenant-level data isolation.
- Prevent unauthorized access to identifiable customer data.
- Audit segment use.
- Audit eligibility-rule changes.
- Minimize unnecessary exposure of personal data.

Permission Example

A Campaign Manager may be permitted to see:

Audience size: 126,500

without being permitted to export:

Individual customer identities.

This should be controlled through RBAC/PBAC.

Key Integrations

CRM / CDP

Segments, customer profile attributes and behavioral data.

Membership & Loyalty

Membership/tier eligibility.

Ticketing Catalogue

Product relationships.

Promotion Rule Engine — Board 2

Commercial benefits.

Coupon Engine — Board 3

Code-based eligibility.

Bundle Engine — Boards 5-6

Bundle targeting.

B2B / Reseller

Partner/account eligibility.

POS & Digital Channels

Context and channel.

Payment Orchestration

Payment-method eligibility.

Marketing / Communications

Campaign audience activation.

Reporting / AI

Performance and predictive targeting.

Important Architecture Principle

We should preserve this sequence:

WHO IS THE CUSTOMER / WHAT IS THE CONTEXT?

↓

Board 7 — Targeting & Eligibility

Does this customer qualify?



Board 2 — Promotion Rule Studio

What commercial benefit applies?



Board 8 — Stacking & Conflict Engine

If several offers qualify, which ones are actually applied?



Pricing Engine

What is the final commercial price?

This is much cleaner than putting all four responsibilities into one promotion rule engine.

Matrix Coverage — Board 7

Board 7 directly or substantially supports matrix requirements involving:

- Customer/guest eligibility and targeted discounts.
- Membership and loyalty eligibility.
- Customer/account-specific promotion conditions.
- Channel/location/partner restrictions.
- Transaction and behavioral targeting.
- Payment-method eligibility.
- Promotion conditions based on guest type, membership status, loyalty tier, channel, season, location, and campaign.
- AI-generated promotion audiences and recommendations.
- Personalized upsell/cross-sell using customer profile and historical behavior.
- Consistent targeting across B2C, B2B, POS, kiosk, mobile, and other supported channels.

This board also establishes the audience intelligence foundation required later by the separate **Upsell, Cross-sell & Recommendation Engine module**.

Acceptance Criteria

Board 7 shall be complete when:

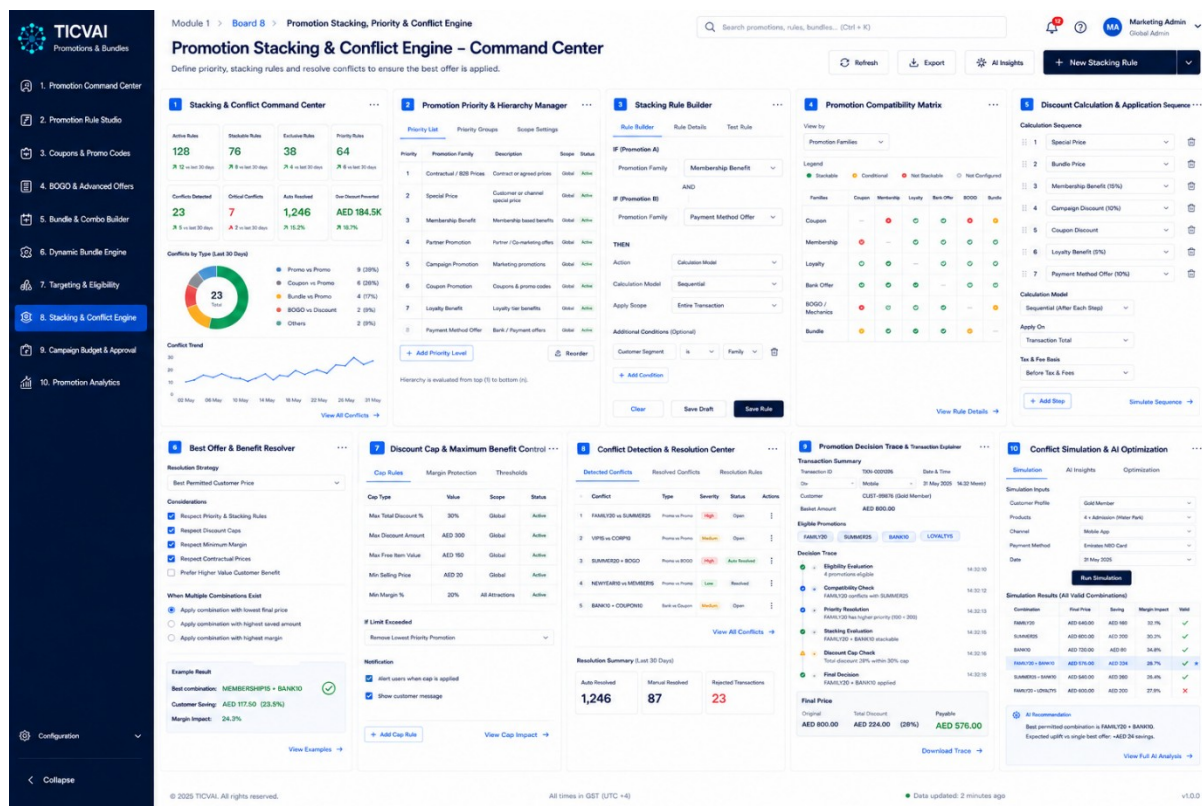
1. Authorized users can create eligibility rules without development.
2. Rules support AND/OR/NOT and nested conditions.
3. Rules can explicitly include and exclude audiences.
4. CRM segments can be consumed without recreating CRM segmentation unnecessarily.
5. Membership status and tier can determine eligibility.
6. Loyalty tier and permitted loyalty attributes can determine eligibility.

7. Guest categories can determine eligibility.
8. Historical purchases can determine eligibility.
9. Visit history can determine eligibility.
10. Previous promotion redemption can determine eligibility.
11. Transaction value and frequency can determine eligibility.
12. Channel can determine eligibility.
13. Venue/location can determine eligibility.
14. Date, time, season, and campaign context can determine eligibility.
15. Partner and B2B account attributes can determine eligibility.
16. Payment-method eligibility can be consumed securely from Payment Orchestration.
17. Administrators can preview audience size before activation.
18. Audience funnels explain how the final audience was derived.
19. Individual profiles/test scenarios can explain why eligibility passed or failed.
20. Frequency caps and cooling-off periods can be configured.
21. Customer and campaign exclusions can be configured.
22. Eligibility conflicts are identified before the stacking engine.
23. AI can discover new potential audiences.
24. AI can predict conversion, revenue, and promotion response.
25. AI recommendations are explainable.
26. AI-generated segments require governed activation.
27. Customer data visibility respects RBAC/PBAC.
28. Tenant/customer data isolation is enforced.
29. Targeting and eligibility changes are versioned and audited.
30. The engine exposes eligibility consistently to all authorized TICVAI sales and recommendation channels.

Board 7 is therefore locked as: *Promotion Targeting, Eligibility & Segmentation Engine — 10 backend screens.*

Module 1 — Promotions & Bundles Management

Board 8 — Promotion Stacking, Priority & Conflict Engine



Board Objective

Board 8 shall provide TICVAI with the centralized **Promotion Decision Engine** responsible for determining what happens when **multiple promotions, discounts, coupons, bundles, loyalty benefits, membership benefits, payment offers, BOGO mechanics, partner offers, or special prices** qualify for the same transaction.

This is one of the most important boards in the Promotions module because the previous boards may all produce valid offers simultaneously.

For example, a guest could qualify for:

Annual Pass Holder — 15% discount
 SUMMER20 coupon — 20% discount
 Emirates NBD card — 10% discount
 Buy 4 Pay 3 — promotional mechanic
 Loyalty Gold — 5% benefit

TICVAI must determine automatically:

Can they combine? In what order? Which promotion wins? What is the maximum permitted benefit?

The matrix explicitly requires configurable promotion hierarchy, stacking and conflict resolution, including scenarios where promotions are **combined, mutually exclusive, or prioritized**, with clear explanation of which promotion was applied and why.

Board 8 shall contain **10 backend screens**.

Screen 1 — Stacking & Conflict Command Center

Purpose

Provide centralized operational visibility into promotion interactions across TICVAI.

KPI Cards

- Active Promotion Rules
- Stackable Promotions
- Exclusive Promotions
- Priority Rules
- Conflicts Detected
- Critical Conflicts
- Auto-Resolved Conflicts
- Transactions with Multiple Offers
- Average Promotions per Transaction
- Discount Exposure
- Prevented Over-Discount
- Revenue Protected

Conflict Categories

- Promotion vs Promotion
- Coupon vs Promotion
- Coupon vs Coupon
- BOGO vs Discount
- Bundle vs Promotion
- Membership vs Promotion
- Loyalty vs Promotion
- Bank Offer vs Promotion
- Partner Offer vs Promotion
- Special Price vs Promotion

Conflict Severity

- Information
- Warning
- High
- Critical

AI Insights

Example:

Conflict Alert: FAMILY20 and SUMMER25 currently overlap across 14 products and three channels.

Revenue Protection: Priority rules prevented AED 184,500 of unintended stacked discounts during the previous 30 days.

Screen 2 — Promotion Priority & Hierarchy Manager

Purpose

Define the relative priority of different promotion families and individual offers.

Example Hierarchy

1. Contractual / B2B Price
2. Special Price
3. Membership Benefit
4. Partner Promotion
5. Campaign Promotion
6. Coupon
7. Loyalty Benefit
8. Payment Method Offer

The hierarchy must be configurable rather than hard-coded.

Priority Configuration

For each promotion:

- Promotion family
- Promotion
- Priority number
- Priority group
- Effective dates
- Channel
- Venue
- Customer segment
- Override behavior

Example

FAMILY20

Priority: **200**

SUMMER10

Priority: **300**

If mutually exclusive:

FAMILY20 wins.

Hierarchy Scope

Hierarchy may differ by:

- Tenant
- Business entity
- Venue
- Channel
- Product category
- Customer type
- Campaign

Screen 3 — Promotion Stacking Rule Builder

Purpose

Define which promotions can be combined.

Stacking Models

Support:

Fully Stackable

Promotion A + Promotion B.

Non-Stackable

Only one promotion can apply.

Conditional Stacking

Can stack only under specific conditions.

Category Stacking

One promotion from each promotion category.

Maximum N Promotions

Example:

Maximum two promotions per transaction.

Rule Example

Membership Discount
CAN STACK WITH
Bank Offer

but

Membership Discount
CANNOT STACK WITH
Coupon Discount.

No-Code Rule

IF

Promotion Type = Membership

AND

Second Promotion Type = Bank Offer

THEN

Allow Stack.

Scope

Rules may apply to:

- Entire transaction
- Product
- Product category
- Individual ticket
- Bundle component
- Customer
- Channel

Screen 4 — Promotion Exclusion & Compatibility Matrix

Purpose

Provide administrators with a visual matrix showing which promotion categories can interact.

Example

	Coupon	Membership	Loyalty	Bank	BOGO
Coupon	✗	✗	✓	✓	✗
Membership	✗	—	✓	✓	✓
Loyalty	✓	✓	—	✓	✓
Bank	✓	✓	✓	—	✓
BOGO	✗	✓	✓	✓	—

Status

Use:

- Allowed
- Not Allowed
- Conditional
- Priority Based
- Not Configured

Drill Down

Clicking a relationship should show:

- Applicable products
- Applicable venues
- Channels
- Customer segments
- Rule
- Priority
- Effective dates
- Exceptions

Conflict Detection

The system should identify missing combinations.

Example:

16 active promotion relationships currently have no explicit compatibility rule.

Screen 5 — Discount Calculation & Application Sequence

Purpose

Control the mathematical order in which multiple permitted benefits are calculated.

This is critical because:

20% + 10%

can produce different results depending on calculation method.

Example

Original price:

AED 100

Sequential

20% discount:

AED 100 → AED 80

Then 10%:

AED 80 → AED 72

Final = AED 72

Additive

20% + 10% = 30%

AED 100 → AED 70

Final = AED 70

Supported Calculation Models

- Sequential
- Additive percentage
- Fixed then percentage
- Percentage then fixed
- Best price only
- Highest-value discount only
- Lowest-price result
- Priority order

Application Sequence

Example:

1. Special price
2. Bundle price
3. Membership benefit
4. Campaign discount
5. Coupon
6. Loyalty
7. Payment offer

Tax/Fee Basis

Configure whether promotion applies:

- Before tax

- After tax
- Before fee
- After fee
- Product only
- Transaction total

Actual tax calculation remains governed by TICVAI's pricing/finance rules.

Screen 6 — Best Offer & Customer Benefit Resolver

Purpose

Determine which promotion or combination gives the guest the correct/best permitted commercial outcome.

Resolution Strategies

Highest Priority

Use the highest-ranked promotion.

Best Customer Price

Apply the eligible combination producing the lowest permitted price.

Highest Margin

Use the commercially permitted offer providing the strongest margin.

Campaign Preference

Favor a strategically selected campaign.

Contractual Priority

Honor contractual/customer-specific pricing first.

Example

Eligible offers:

Offer	Saving
FAMILY15	AED 75
SUMMER20	AED 100
BANK10	AED 50
FAMILY15 + BANK10	AED 117.50

Configured strategy:

Best permitted customer price

Result:

FAMILY15 + BANK10

Saving:

AED 117.50

Customer Transparency

The sales channel should be able to show:

Best available offer applied automatically.

Screen 7 — Discount Cap & Maximum Benefit Controller

Purpose

Prevent stacked promotions from exceeding financial or contractual limits.

Controls

Configure:

- Maximum total discount %
- Maximum total discount amount
- Minimum transaction value
- Minimum product price
- Minimum margin
- Maximum free-item value
- Maximum promotional benefit
- Maximum promotions per basket

Example

Eligible stack:

Membership 15%

•

Bank 10%

•

Coupon 20%

Potential combined benefit:

45%

Configured maximum:

30%

TICVAI shall automatically resolve the transaction according to configured rules.

Cap Scope

Limits can exist at:

- Tenant
- Venue
- Product
- Category
- Campaign
- Customer
- Partner
- Transaction
- Channel

Cap Actions

When limit is exceeded:

- Reduce discount
 - Remove lowest-priority promotion
 - Remove highest-cost promotion
 - Use best permitted combination
 - Require approval
 - Reject combination
-

Screen 8 — Conflict Detection & Resolution Center

Purpose

Detect promotion conflicts during both configuration and runtime.

Configuration-Time Conflicts

Detect:

- Same product
- Same audience
- Same channel
- Same validity
- Incompatible promotions
- Missing hierarchy
- Missing stacking rule
- Discount-cap breach
- Circular dependency

Runtime Conflicts

Example:

Customer qualifies for:

- FAMILY20
- VIP15
- SUMMER25
- BANK10

The engine evaluates:

Eligibility

↓

Compatibility

↓

Priority

↓

Calculation sequence

↓

Discount cap

↓

Final result

Conflict Resolution

Support:

- Automatic
- Rule-based
- Best price
- Priority
- Manual intervention
- Reject transaction where necessary

Resolution Explanation

Example:

SUMMER25 was not applied because FAMILY20 has higher priority and the two promotions are mutually exclusive.

Screen 9 — Promotion Decision Trace & Transaction Explainer

Purpose

Provide complete explainability of how TICVAI arrived at the final promotional price.

This screen will be extremely important for:

- Customer service
- Finance
- Operations
- Audit
- Partner disputes
- Technical support

Transaction Example

Original Basket

AED 800

Eligible Offers

FAMILY20 ✓

SUMMER25 ✓

BANK10 ✓

LOYALTY5 ✓

Decision Trace

Step 1

FAMILY20 eligible.

Step 2

SUMMER25 eligible but conflicts with FAMILY20.

Step 3

FAMILY20 priority = 100.

SUMMER25 priority = 200.

FAMILY20 retained.

Step 4

BANK10 stackable with FAMILY20.

Step 5

LOYALTY5 not stackable with Bank offer.

Step 6

Best permitted combination calculated.

Final Result**Original:**

AED 800

Family discount:

–AED 160

Bank discount:

–AED 64

Final:

AED 576

Total Saving:

AED 224 / 28%

Explanation

FAMILY20 + BANK10 applied because they provide the best permitted combination under the configured stacking and 30% maximum-discount policy.

Screen 10 — Conflict Simulation & AI Optimization

Purpose

Test promotion interaction scenarios before publishing campaigns.

Simulation Inputs**Administrators can select:**

- Customer segment
- Membership
- Loyalty
- Products
- Basket
- Coupon
- Channel
- Venue
- Payment method
- Date/time

Example Scenario**Customer:**

Gold Member

Basket:

4 × Admission**Coupon:****SUMMER20****Payment:****Bank ABC****Eligible Promotions**

- Gold Member 15%
- Buy 4 Pay 3
- SUMMER20
- Bank ABC 10%

Simulation

TICVAI evaluates every permitted combination.

Combination	Saving	Margin	Valid
Member	AED 120	36%	✓
BOGO	AED 200	29%	✓
Coupon	AED 160	33%	✓
Member + Bank	AED 188	31%	✓
BOGO + Coupon	AED 328	13%	✗
BOGO + Bank	AED 260	24%	✓

Recommended Result**BOGO + Bank****Customer Saving:****AED 260****Margin:****24%****AI Requirements for Board 8**

AI should assist administrators in identifying complex commercial interactions that may be difficult to detect manually.

AI shall detect:

- Promotion collisions
- Excessive stacking

- Unexpected discount exposure
- Margin erosion
- Conflicting eligibility
- Circular rules
- Redundant promotions
- Cannibalization
- Unusual stacking patterns

AI shall recommend:

- Priority changes
- Stacking restrictions
- Maximum benefit caps
- Better calculation sequence
- Campaign exclusions
- Promotion consolidation
- Safer commercial combinations

Example AI Alert

Margin Risk Detected

SUMMER20 + Gold Membership + Bank10 can currently stack across 14 products.

In 8 products, the resulting margin falls below the configured 20% threshold.

Recommended action: Keep Gold Membership + Bank10 stackable but make SUMMER20 mutually exclusive.

Estimated protected margin: **AED 284K/month.**

AI Simulation

AI should also run large-scale simulations using historical basket patterns to identify conflicts **before a campaign is activated.**

Important Runtime Architecture

Board 8 should sit very close to TICVAI's Pricing Engine.

The transaction sequence should be:

Basket

↓

Board 7 — Targeting & Eligibility

Who qualifies?

↓

Boards 2-4 — Promotion Mechanics

Which promotions qualify and what benefits do they generate?

↓

Board 8 — Stacking & Conflict Engine

Which benefits can actually coexist?

↓

Discount Guardrails

Are commercial limits respected?

↓

Pricing Engine

Calculate authoritative final price.

↓

Payment

Collect final amount.

This means the sales channels should **never independently calculate stacking logic**.

B2C, POS, B2B, Mobile App, Kiosk, Call Center and API should all receive the result from the same centralized engine.

Roles & Governance

Typical roles:

- Marketing Administrator
- Campaign Manager
- Commercial Manager
- Revenue Manager
- Finance
- B2B Manager
- System Administrator
- Approver
- Auditor

Sensitive Permissions

Separate permissions should control:

- Change priority
- Allow stacking
- Create exclusions
- Change calculation sequence
- Change discount cap
- Override margin floor
- Create emergency rule
- Activate hierarchy changes

High-risk changes should support approval workflows.

Key Integrations

Board 2 — Promotion Rule Studio

Promotion calculations.

Board 3 — Coupons

Coupon benefits.

Board 4 — Advanced Promotions

BOGO and reward mechanics.

Boards 5-6 — Bundles

Bundle price and component interactions.

Board 7 — Targeting

Eligibility.

Membership & Loyalty

Member benefits.

Pricing / Dynamic Pricing

Authoritative pricing.

Payment

Payment-method offers.

B2B / Reseller

Contractual prices and partner promotions.

CRM

Customer segmentation.

Finance

Margin and financial controls.

Reporting / AI

Conflict analytics and optimization.

Matrix Coverage — Board 8

Board 8 directly addresses one of the most explicit areas of the matrix:

3.6.35 — Promotion hierarchy and conflict resolution, including:

- Combine promotions.
- Make promotions mutually exclusive.
- Prioritize promotions.
- Automatically determine applicable promotion.
- Clearly indicate which promotion was applied and why.

It also provides the decision layer required to safely execute:

- Percentage/value discounts.
- Coupons.

- BOGO.
- Volume discounts.
- Membership offers.
- Loyalty offers.
- Bank/payment offers.
- Bundles.
- Partner promotions.
- Special prices.
- Cart-level promotions.

The board therefore does not duplicate those promotion mechanics; it **orchestrates their interaction**.

Acceptance Criteria

Board 8 shall be considered complete when:

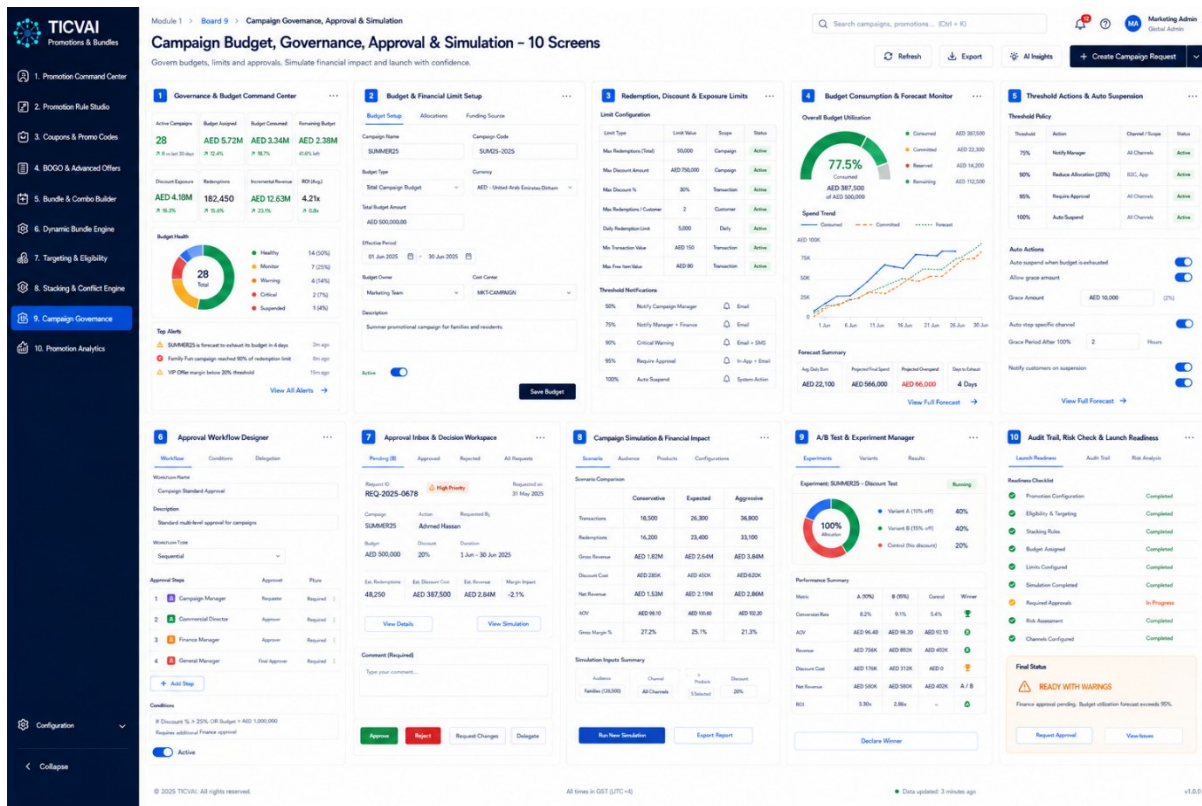
1. Promotion priorities can be configured.
2. Priority can differ by tenant, venue, channel, product, and segment.
3. Promotions can be configured as stackable.
4. Promotions can be configured as mutually exclusive.
5. Conditional stacking is supported.
6. Maximum number of simultaneous promotions can be configured.
7. Promotion-category compatibility can be configured.
8. Individual promotion exclusions can be configured.
9. Calculation order can be configured.
10. Sequential and additive discount calculations are supported where permitted.
11. Fixed and percentage discounts can interact according to configured sequence.
12. Best-price resolution is supported.
13. Highest-priority resolution is supported.
14. Contractual pricing can be protected.
15. Maximum total discount can be enforced.
16. Minimum selling price can be protected.
17. Minimum margin can be protected.
18. Conflicts can be detected before campaign activation.
19. Runtime conflicts can be resolved automatically.
20. TICVAI can explain why a promotion was applied.
21. TICVAI can explain why another promotion was rejected.
22. Complete transaction-level promotion decision traces are retained.
23. Users can simulate multiple promotions before activation.
24. Simulation shows customer saving and margin impact.
25. AI can detect risky promotion combinations.
26. AI can recommend hierarchy and stacking changes.
27. AI can simulate historical basket behavior.

- 28. High-risk configuration changes can require approval.
- 29. All stacking/hierarchy changes are versioned and audited.
- 30. Every TICVAI sales channel consumes the **same centralized promotion decision**, ensuring that the same basket and customer conditions produce the same governed promotional outcome across B2C, B2B, POS, Kiosk, Mobile App, Call Center, and API.

Board 8 is locked as: *Promotion Stacking, Priority & Conflict Engine — 10 backend screens.*

Module 1 — Promotions & Bundles Management

Board 9 — Campaign Budget, Governance, Approval & Simulation



Board Objective

Board 9 shall provide TICVAI with the centralized **financial governance and approval control layer for promotions and bundles**.

Boards 1–8 define the promotion, its commercial rules, audience, bundles, advanced mechanics, and conflict behavior. Board 9 determines:

“Are we financially and operationally authorized to launch this campaign, how much commercial exposure are we willing to accept, who must approve it, and when should TICVAI automatically intervene?”

The matrix specifically requires campaign-level **budgets, redemption limits, discount caps, automated suspension, multi-level approval workflows, audit tracking, campaign simulation, financial-impact forecasting, and A/B testing before or during activation**.

Board 9 shall contain **10 backend screens**.

Screen 1 — Campaign Governance & Budget Command Center

Purpose

Provide executives, Marketing, Commercial, Revenue, and Finance with one centralized view of the financial and governance status of promotional campaigns.

KPI Cards

- Active Campaigns

- Campaign Budget
- Budget Consumed
- Remaining Budget
- Discount Exposure
- Redemption Value
- Revenue Generated
- Incremental Revenue
- Campaign ROI
- Campaigns Near Budget Limit
- Pending Approvals
- Suspended Campaigns

Budget Health

Classify campaigns as:

- Healthy
- Monitor
- Warning
- Critical
- Budget Exhausted
- Suspended

Example

Summer Family Campaign

Budget: AED 500,000

Consumed: AED 387,500

Remaining: AED 112,500

Utilization: 77.5%

Revenue: AED 2.84M

ROI: 5.3x

Status: HEALTHY

AI Insights

Example:

SUMMER25 is forecast to exhaust its AED 500,000 budget four days before campaign expiry.

Screen 2 — Campaign Budget & Financial Limit Setup

Purpose

Define the financial envelope within which a campaign is permitted to operate.

Budget Types

Support:

- Total campaign budget
- Discount budget
- Reward budget
- Free-product budget

- Partner-funded budget
- Marketing-funded budget
- Venue budget
- Department budget

Configuration

- Campaign
- Budget amount
- Currency
- Effective dates
- Budget owner
- Cost center
- Business entity
- Venue
- Department
- Funding source

Budget Scope

A budget may apply to:

- Entire campaign
- Promotion
- Product
- Venue
- Channel
- Partner
- Customer segment

Example

Campaign Budget

AED 1,000,000

Allocation:

Area	Budget
B2C	AED 500K
Mobile App	AED 200K
B2B	AED 150K
POS	AED 100K
Partner	AED 50K

Shared vs Dedicated Budget

Support:**Shared Pool**

All promotions consume one campaign budget.

Dedicated

Each promotion receives its own allocation.

Screen 3 — Redemption, Discount & Exposure Limit Manager**Purpose**

Define non-budget commercial limits controlling campaign exposure.

Limit Types

- Maximum redemptions
- Maximum discount value
- Maximum discount %
- Maximum reward quantity
- Maximum free tickets
- Maximum free products
- Maximum transactions
- Maximum customers
- Maximum redemptions per customer
- Maximum daily exposure

Example**Campaign:**

SUMMER25

Maximum redemptions:

50,000

Maximum total discount:

AED 750,000

Maximum customer redemption:

2

Daily redemption limit:

5,000

Multi-Level Limits**Support limits at:**

Campaign

↓

Promotion

↓

Channel

↓

Venue



Customer

Threshold Notifications

Configure:

- 50%
- 75%
- 90%
- 95%
- 100%

or custom thresholds.

Screen 4 — Budget Consumption & Forecast Monitor

Purpose

Provide real-time tracking of campaign financial consumption and predict when limits will be reached.

Dashboard

Display:

- Original budget
- Consumed
- Committed
- Reserved
- Remaining
- Forecast final spend
- Daily burn rate

Budget States

Distinguish between:

Consumed

Completed transaction.

Reserved

Promotion benefit temporarily reserved during checkout.

Committed

Transaction authorized but not yet fully settled.

Forecast

Example:

Current budget:

AED 500K

Consumed:

AED 390K

Average daily burn:

AED 22K

Remaining campaign:**8 days****Projected spend:****AED 566K****Projected overspend:****AED 66K****AI Forecast**

At the current redemption velocity, the campaign will exceed its budget by approximately 13.2%.

Screen 5 — Threshold Actions & Automatic Suspension

Purpose

Define what TICVAI should do as financial or redemption thresholds are approached or exceeded.

This is explicitly required by the matrix.

Example Policy**At 75%**

Notify Campaign Manager.

At 90%

Notify Campaign Manager + Finance.

At 95%

Critical warning.

At 100%

Automatically suspend promotion.

Available Actions

- Notify
- Warn
- Require approval
- Reduce allocation
- Stop specific channel
- Stop partner
- Stop promotion
- Stop campaign
- Allow grace amount
- Continue with executive authorization

Grace Configuration**Example:****Budget:**

AED 500,000

Allowed grace:

2% / AED 10,000

Hard stop:

AED 510,000

Fail-Safe

TICVAI must ensure that concurrent transactions cannot cause uncontrolled overspending simply because multiple checkouts reach the limit simultaneously.

The runtime budget service therefore needs atomic/reservation-based consumption control.

Screen 6 — Campaign Approval Workflow Designer

Purpose

Configure multi-level approval workflows for promotions and campaigns.

The matrix explicitly requires configurable multi-level approval for promotion **creation, modification, activation, and deactivation**.

Workflow Example

Campaign Manager

↓

Commercial Director

↓

Finance

↓

General Manager

↓

Approved

Approval Conditions

Workflow may depend on:

- Discount %
- Campaign budget
- Margin
- Promotion type
- Venue
- Partner
- Channel
- Free-product value
- Campaign duration
- Financial exposure

Conditional Approval Example

If:

Discount ≤10%

→ Marketing Manager.

If:

Discount 10–25%

→ Commercial Director.

If:

Discount >25%

→ Commercial Director + Finance.

If:

Budget >AED 1M

→ CFO/GM.

Workflow Configuration

Support:

- Sequential approval
 - Parallel approval
 - Conditional approval
 - Mandatory approval
 - Optional review
 - Delegation
 - Escalation
-

Screen 7 — Approval Inbox & Decision Workspace

Purpose

Provide approvers with enough commercial information to make an informed decision without navigating through every configuration screen.

Approval Card

Show:

- Campaign
- Promotion
- Requested by
- Request date
- Requested action
- Discount
- Budget
- Estimated redemptions
- Estimated revenue
- Margin impact
- Customer reach
- Risk level
- AI forecast

Change Comparison

For modifications:

Current

VS

Proposed

Example:

Discount:

15% → **25%**

Budget:

AED 500K → **AED 750K**

Duration:

14 days → **30 days**

Approver Actions

- Approve
- Reject
- Return for Change
- Request Information
- Delegate

Comments

Approvers may provide mandatory decision comments.

Approval History

Show the complete chain:

Marketing ✓

Commercial ✓

Finance Pending

GM Not Started

Screen 8 — Campaign Financial & Commercial Simulator

Purpose

Simulate the likely financial result of a campaign before activation.

This is a major matrix requirement.

Simulation Inputs

- Audience
- Products
- Promotion
- Discount
- Duration
- Channels
- Historical conversion
- Expected traffic
- Redemption limit
- Budget

Simulation Outputs

Show:

- Eligible audience
- Expected transactions
- Expected redemptions
- Gross revenue
- Discount cost
- Net revenue
- Incremental revenue
- Average order value
- Gross margin
- Margin impact
- Expected budget consumption
- ROI

Scenario Comparison**Support:****Conservative****Expected****Aggressive****Example:**

Scenario	Revenue	Discount	Margin	ROI
Conservative	AED 1.8M	AED 320K	27%	3.6x
Expected	AED 2.6M	AED 450K	25%	4.8x
Aggressive	AED 3.4M	AED 620K	21%	5.1x

Historical Replay

Where data is available, TICVAI should be able to apply proposed campaign rules against historical transactions to estimate what would have occurred.

Screen 9 — Campaign Experiment & A/B Test Manager**Purpose**

Allow TICVAI to test campaign variants and determine which commercial strategy performs better.

The matrix explicitly requires A/B testing of promotion variants, including discount levels, validity periods, bundles, and target segments.

Experiment Example**Variant A**

10% discount.

Variant B

15% discount.

Variant C

Free F&B voucher.

Audience Allocation

- A — 40%
- B — 40%
- Control — 20%

Experiment Dimensions**Test:**

- Discount %
- Discount value
- Promotion type
- Bundle
- Reward
- Audience
- Channel
- Validity
- Message/offer
- Timing

Performance**Measure:**

- Conversion
- Revenue
- AOV
- Redemption
- Discount cost
- Margin
- Incremental revenue
- ROI

Winner Selection**Support:**

- Manual winner
- Rule-based winner
- AI recommendation

Example

Variant B produced +8.7% conversion but Variant A generated 4.2% higher margin. The user decides whether conversion or profitability is the primary objective.

Screen 10 — Governance Audit, AI Risk & Launch Readiness

Purpose

Provide the final governance checkpoint before a campaign is allowed to go live.

Launch Readiness Checklist

Promotion Configuration ✓

Eligibility ✓

Stacking Rules ✓

Budget ✓

Redemption Limits ✓

Margin Guardrail ✓

Simulation Completed ✓

Required Approval ✓

Channel Publication ✓

Audit Requirements ✓

Final Status

Possible outcomes:

READY TO LAUNCH

READY WITH WARNINGS

BLOCKED

Blocking Issues

Examples:

- Finance approval missing
- Budget not assigned
- Discount exceeds authorization
- Margin below threshold
- Conflict unresolved
- Simulation not completed
- Required product unavailable

Audit

Record:

- Budget creation/change
- Limit changes
- Approval submissions
- Approval decisions
- Overrides
- Automatic suspension
- Reactivation
- Simulation results
- Experiment changes
- Campaign launch

AI Requirements for Board 9

AI should act as the **financial and governance co-pilot** for promotion campaigns.

AI shall forecast:

- Redemption volume
- Discount exposure
- Budget consumption
- Budget exhaustion date
- Revenue
- Incremental revenue
- AOV
- Margin
- ROI

AI shall detect:

- Unusual budget consumption
- Campaign overspend risk
- Unrealistic redemption assumptions
- Excessive discount
- Poor margin
- Weak ROI
- Cannibalization risk
- Approval risk
- Underperforming experiments

AI shall recommend:

- Campaign budget
- Discount cap
- Redemption limit
- Campaign duration
- Audience size
- Budget allocation by channel
- Early suspension
- Campaign extension
- Winning experiment variant

Example

AI Budget Recommendation

Current proposed budget: **AED 750,000**

Forecast required budget: **AED 610,000**

Recommended initial budget: **AED 625,000**

Maintain AED 125,000 as reserve and release it only if conversion remains above 8.5%.

This is much stronger than simply saying "the campaign may perform well."

Human Approval & AI Authority

For TICVAI, I recommend three AI action levels:

Level 1 — Advisory

AI recommends.

Human decides.

Level 2 — Governed Automation

AI may perform a pre-approved action.

Example:

At 90% budget utilization, reduce B2C daily allocation by 20%.

Level 3 — Financial Protection

The system automatically performs mandatory safeguards.

Example:

Hard campaign budget reached → stop further qualifying redemptions.

AI should **never independently increase an approved campaign budget or commercial authorization ceiling.**

That must remain a governed human decision.

Key Integrations

Board 1 — Promotion Command Center

Campaign operational status.

Boards 2-4 — Promotion Mechanics

Discount/reward exposure.

Boards 5-6 — Bundles

Bundle financial impact.

Board 7 — Targeting

Audience size.

Board 8 — Stacking

Combined discount exposure.

Pricing / Dynamic Pricing

Price and margin.

CRM

Audience and historical response.

Finance

Budget, accounting and profitability.

Payment

Actual transaction values.

Reporting / AI

Forecasts and campaign analytics.

RBAC/PBAC

Approval authority.

Important Architecture Principle

Board 9 is the **governance gate between commercial configuration and live execution**.

The flow should become:

Boards 2-8

Configure commercial behavior

↓

Board 9 — Governance

Budget → Simulation → Risk → Approval → Launch Readiness

↓

Board 1

Schedule / Activate / Monitor

↓

Sales Channels

Execute campaign

This prevents a Marketing user from configuring a 50% promotion and immediately pushing it live without financial controls.

Matrix Coverage — Board 9

Board 9 directly addresses the matrix requirements around:

3.6.36 — campaign-level budgets, redemption limits, discount caps, thresholds, alerts, and automatic suspension.

3.6.37 — simulation of redemption, discount exposure, revenue impact, margin impact, and financial performance before activation.

3.6.38 — configurable multi-level approval workflows and audit tracking for creation, modification, activation, and deactivation.

3.6.39 — campaign variant testing/A-B testing.

It also provides governance for:

- Flash sales
- Coupons
- Volume discounts
- BOGO
- Bundles
- Payment-method promotions
- Partner campaigns
- Membership offers

- AI-recommended promotions
- Promotion stacking.

This means Board 9 should **not recreate these promotion mechanics**. It governs their financial exposure and authorization.

Acceptance Criteria

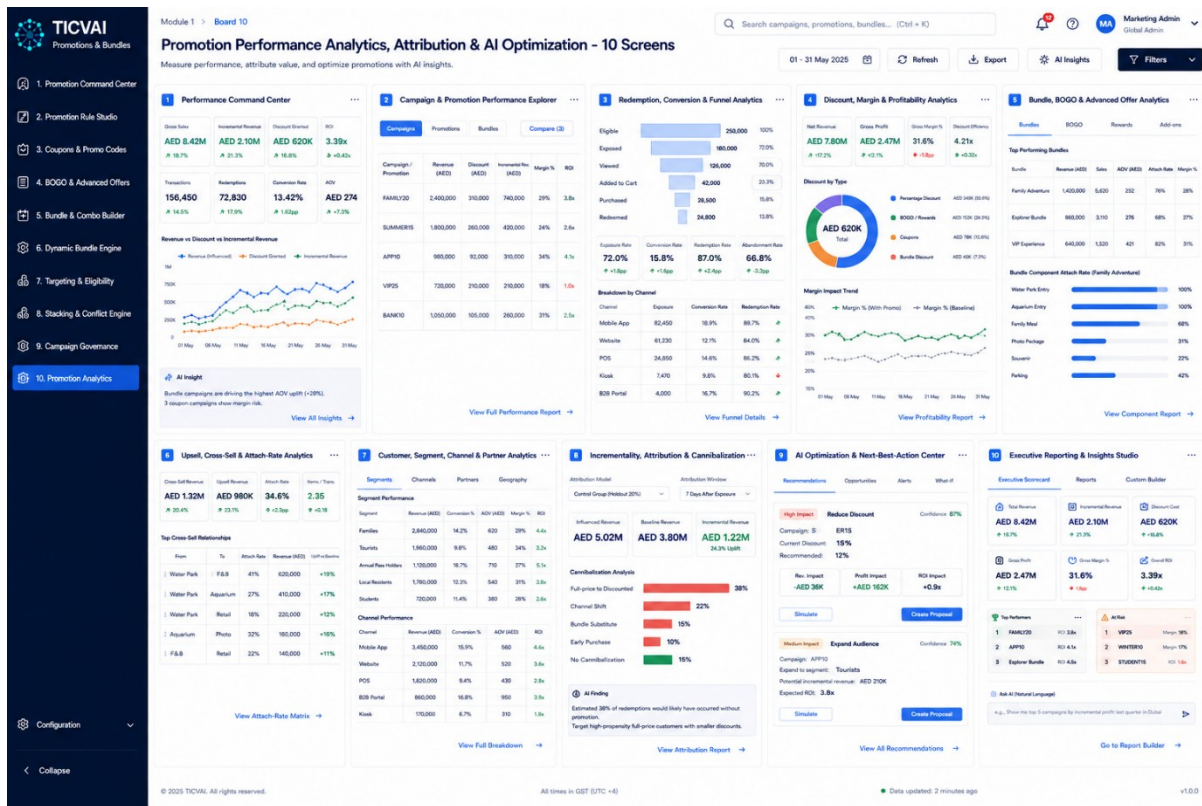
Board 9 shall be considered complete when:

1. Campaign budgets can be configured.
2. Multiple budget types are supported.
3. Budgets can be allocated by venue, channel, promotion, or partner.
4. Shared and dedicated budget pools are supported.
5. Redemption limits can be configured.
6. Discount-exposure limits can be configured.
7. Customer-level redemption limits can be configured.
8. Threshold alerts can be configured.
9. Budget consumption is monitored in real time.
10. Reserved, committed, and consumed exposure can be distinguished.
11. TICVAI forecasts campaign budget exhaustion.
12. Automatic actions can be triggered at thresholds.
13. Hard budget limits prevent uncontrolled overspending.
14. Concurrent transactions cannot bypass the budget ceiling.
15. Multi-level approval workflows are configurable.
16. Approval workflows can depend on discount, budget, margin, and risk.
17. Sequential and parallel approvals are supported.
18. Approvers can compare proposed changes against current configuration.
19. Approval decisions and comments are audited.
20. Campaign financial performance can be simulated before activation.
21. Conservative, expected, and aggressive scenarios are supported.
22. Historical transaction replay can be used where data exists.
23. A/B campaign variants can be configured.
24. Audience allocation between variants and control groups is supported.
25. Conversion, revenue, AOV, discount cost, margin, and ROI can be compared.
26. AI can recommend a winning experiment but governed users retain decision authority.
27. A final launch-readiness check validates all required commercial and governance controls.
28. Missing mandatory approval can block activation.
29. Missing budget or unresolved financial risk can block activation.
30. All budget, approval, simulation, experiment, override, and suspension activities are fully versioned and audited.

Board 9 is therefore locked as: Campaign Budget, Governance, Approval & Simulation — 10 backend screens.

Module 1 — Promotions & Bundles Management

Board 10 — Promotion Performance Analytics, Attribution & AI Optimization



Board Objective

Board 10 shall be the **performance intelligence and optimization layer** for the complete Promotions & Bundles module.

Boards 1-9 answer:

What campaigns exist → What rules apply → Who qualifies → What offers/bundles are created → How conflicts are resolved → What budget and approvals govern them.

Board 10 answers:

“What actually happened, did the promotion create incremental business value, which promotions and bundles are performing best, what is hurting margin, and what should we do next?”

This board must go beyond basic sales reporting. TICVAI should distinguish between **revenue generated through a promotion** and the estimated **incremental revenue caused by the promotion**, while also measuring discount cost, margin, redemption, conversion, cross-sell, upsell, bundle performance, partner/channel performance, and campaign ROI.

Board 10 shall contain **10 backend screens**.

Screen 1 — Promotion Performance Command Center

Purpose

Provide executives, Marketing, Commercial, Revenue, and Finance with the overall performance of promotions and bundles.

Global Filters

- Date range
- Business entity
- Venue
- Attraction
- Campaign
- Promotion
- Bundle
- Channel
- Customer segment
- Product
- Partner
- Market
- Currency

KPI Cards

- Gross Sales
- Promotion-Influenced Revenue
- Estimated Incremental Revenue
- Discount Granted
- Net Revenue
- Gross Margin
- Promotion Cost
- ROI
- Transactions
- Redemptions
- Conversion Rate
- Average Order Value
- Revenue per Redemption
- Active Campaigns

Performance Trend

Show:

Revenue vs Discount Cost vs Incremental Revenue

over time.

Executive AI Summary

Example:

Promotions generated AED 8.4M in influenced revenue this month. Estimated incremental revenue was AED 2.1M, with AED 620K of discount exposure. Bundle campaigns produced

the strongest AOV uplift, while three coupon campaigns are showing potential margin erosion.

Screen 2 — Campaign & Promotion Performance Explorer

Purpose

Allow users to compare every promotion and campaign using consistent commercial KPIs.

Performance Table

Campaign	Revenue	Discount	Incremental Rev.	Margin	ROI	Conversion
FAMILY20	AED 2.4M	310K	740K	29%	3.8x	11.8%
SUMMER15	AED 1.8M	260K	420K	24%	2.6x	9.4%
APP10	AED 980K	92K	310K	34%	4.1x	13.2%

Compare Promotions

Users can select multiple campaigns and compare:

- Revenue
- Transactions
- Units
- Redemptions
- Discount
- Margin
- Conversion
- AOV
- ROI
- Customer acquisition
- Repeat purchase

Performance Status

AI/system classification:

- Excellent
- Healthy
- Monitor
- Underperforming
- Margin Risk
- Critical

Drill Down

Campaign → Promotion → Product → Customer → Transaction

subject to user permissions.

Screen 3 — Redemption, Conversion & Funnel Analytics

Purpose

Measure how effectively offers move customers from exposure to purchase and redemption.

Promotion Funnel

Eligible

↓

Exposed

↓

Viewed

↓

Clicked / Selected

↓

Added to Cart

↓

Applied

↓

Purchased

↓

Redeemed

Example

Eligible: **250,000**

Exposed: **180,000**

Viewed: **126,000**

Added to Cart: **42,000**

Purchased: **28,500**

Redeemed: **24,800**

KPIs

- Exposure rate
- Engagement rate
- Cart rate
- Conversion rate
- Redemption rate
- Abandonment
- Unused promotion rate
- Expired benefit rate

Breakdown

Analyze by:

- Channel
- Customer segment

- Membership tier
- Product
- Venue
- Campaign
- Promotion type
- Day/time
- Device
- Partner

This screen should consume exposure/click events only where the relevant channel actually captures them.

Screen 4 — Discount, Margin & Profitability Analytics

Purpose

Determine whether promotions are commercially profitable rather than merely generating sales.

Core Metrics

- Gross Revenue
- Discount Value
- Net Revenue
- Product Cost
- Promotion Cost
- Gross Profit
- Gross Margin %
- Margin Change
- Revenue Uplift
- Profit Uplift
- ROI

Discount Efficiency

Calculate indicators such as:

Incremental Revenue / Discount Cost

Example:

Discount cost:

AED 100,000

Estimated incremental revenue:

AED 420,000

Efficiency:

4.2x

Margin Risk Matrix

Plot promotions by:

Revenue Impact

versus

Margin Impact

to identify:

- High revenue / high margin
- High revenue / low margin
- Low revenue / high margin
- Low revenue / low margin

AI Warning

Campaign VIP25 increased conversion by 12.4%, but gross margin declined from 31% to 18%. The additional conversion is not currently offsetting the discount cost.

Screen 5 — Bundle, BOGO & Advanced Offer Analytics

Purpose

Measure performance specifically for the commercial mechanics created in Boards 4–6.

Bundle KPIs

- Bundle Sales
- Bundle Revenue
- Bundle Conversion
- Bundle AOV
- Component Attach Rate
- Component Redemption
- Bundle Margin
- Bundle vs Standalone Revenue
- Substitution Rate
- Availability Failure Rate

BOGO Metrics

- BOGO transactions
- Free items issued
- Average reward value
- Incremental units
- Incremental revenue
- Margin impact

Bundle Component Analysis

Example:

Family Adventure Bundle

Water Park — 100%

Aquarium — 100%

Meal A — 68%

Meal B — 22%

Meal C — 10%

Photo Add-on — 31%

Parking — 42%

AI Opportunity

Guests selecting Meal B have a 24% higher Retail attachment rate. Consider testing Meal B as the default recommendation for this bundle.

Screen 6 — Upsell, Cross-Sell & Attach-Rate Analytics

Purpose

Measure whether promotions and bundles successfully increase the customer's basket beyond the original purchase.

This screen is particularly important for the cross-sale metrics you originally raised.

KPIs

- Cross-Sell Revenue
- Upsell Revenue
- Attach Rate
- Items per Transaction
- Revenue per Transaction
- Upgrade Conversion
- Recommended Offer Acceptance
- Incremental Basket Value
- Cross-Category Conversion

Product Relationship Analytics

Example:

Water Park → F&B

Attach Rate: **41%**

Water Park → Retail

Attach Rate: **18%**

Water Park → Aquarium

Attach Rate: **27%**

Aquarium → Photo

Attach Rate: **32%**

Cross-Category Matrix

Analyze:

- Ticket → Ticket
- Ticket → F&B
- Ticket → Retail
- Ticket → Experience
- Ticket → Membership
- F&B → Retail
- Retail → F&B

- Membership → Experience

Important Scope Boundary

Board 10 **reports on cross-sell/upsell performance.**

The separate Recommendation/Upsell intelligence capability determines **what should be recommended to the guest in real time.**

This avoids duplicating the recommendation engine inside Promotions.

Screen 7 — Customer, Segment, Channel & Partner Analytics

Purpose

Determine which audiences and distribution channels respond best to promotions.

Customer/Segment KPIs

- Customers reached
- New customers
- Returning customers
- Conversion
- Revenue
- Discount
- AOV
- Margin
- Repeat purchase
- Redemption

Segment Comparison

Example:

Segment	Conversion	AOV	Margin	ROI
Families	14.2%	AED 620	29%	4.4x
Tourists	9.8%	AED 480	34%	3.2x
Members	18.7%	AED 710	27%	5.1x

Channel Analytics

Compare:

- B2C
- Mobile App
- POS
- Kiosk
- B2B
- Call Center

- OTA
- Reseller
- API

Partner Analytics

Measure:

- Partner revenue
 - Partner redemptions
 - Discount cost
 - Commission
 - Net contribution
 - Conversion
 - Campaign ROI
-

Screen 8 — Incrementality, Attribution & Cannibalization Analysis

Purpose

Answer the difficult question:

“Would the customer have purchased anyway without the promotion?”

Without this screen, a campaign with AED 5M revenue could incorrectly be considered highly successful even if most customers would have purchased at full price.

Attribution Models

Where sufficient data exists, support:

- Direct attribution
- Promotion-code attribution
- Campaign attribution
- Control-group comparison
- A/B test attribution
- Pre/post comparison
- Matched audience analysis
- AI-estimated incrementality

Incrementality Example

Promotion-influenced revenue:

AED 5.0M

Estimated baseline revenue:

AED 3.8M

Estimated incremental revenue:

AED 1.2M

Discount cost:

AED 650K

This gives management a much more realistic view of the promotion's contribution.

Cannibalization Analysis

Detect:

- Full-price customers moving to discounted products
- Standard ticket → discounted ticket
- Higher-margin bundle → lower-margin promotion
- Existing member purchase replaced by unnecessary discount
- Channel migration caused by discounting

AI Finding

Approximately 38% of SUMMER20 redemptions came from customer profiles with high predicted full-price purchase probability. Consider narrowing the audience rather than increasing the discount.

Screen 9 — AI Optimization & Next-Best-Action Center

Purpose

Turn analytics into actionable commercial recommendations.

This should be one of the strongest AI screens in the Promotions module.

AI Recommendation Categories

Promotion

- Increase discount
- Reduce discount
- Change mechanic
- Extend promotion
- End promotion
- Change threshold

Audience

- Expand segment
- Narrow segment
- Exclude segment
- Create lookalike audience

Bundle

- Add component
- Remove component
- Replace component
- Change bundle price

Channel

- Expand channel
- Restrict channel
- Reallocate campaign budget

Timing

- Change day

- Change time
- Extend campaign
- Reduce campaign duration

Recommendation Card

Example:

AI Recommendation — Reduce Discount

Campaign: FAMILY20

Current discount: **20%**

Recommended: **15%**

Reason:

Historical simulations indicate that conversion changes by only –1.8 percentage points at 15%, while margin improves by 5.4 points.

Estimated monthly impact:

Revenue: **–AED 42K**

Gross Profit: **+AED 186K**

Actions

Authorized users can:

- Review
- Accept as Draft
- Simulate
- Send for Approval
- Dismiss
- Snooze

AI should **not directly change a live commercial campaign** unless a separately approved automation policy explicitly allows the action.

Screen 10 — Executive Promotion Intelligence & Reporting Studio

Purpose

Provide executive reporting and configurable analytics output across the complete Promotions & Bundles module.

Executive Scorecard

Show:

Total Promotion Revenue

Estimated Incremental Revenue

Total Discount Cost

Incremental Profit

Promotion ROI

Bundle Revenue

Cross-Sell Revenue

Upsell Revenue

Customer Acquisition**Margin Protected****Top / Bottom Performers****Show:****Top Campaigns****by:**

- Incremental revenue
- Profit
- ROI
- Conversion
- AOV

and

Underperformers**by:**

- Negative margin impact
- Low conversion
- High discount cost
- Low incrementality

Reporting Builder**Users can create reports using:**

- Metric
- Dimension
- Filter
- Date
- Comparison period
- Grouping
- Visualization

Export / Distribution**Subject to permissions:**

- Excel
- CSV
- PDF
- Power BI
- API
- Scheduled report

Natural-Language AI Query**Example:**

“Show me the five promotions that generated the highest incremental profit in Dubai last quarter, excluding membership campaigns.”

TICVAI should return the answer with the supporting data and filters used.

AI Requirements for Board 10

Board 10 should consolidate the analytical AI requirements across the complete Promotions & Bundles module.

AI shall analyze

- Campaign performance
- Promotion performance
- Bundle performance
- Coupon performance
- BOGO performance
- Segment response
- Channel performance
- Partner performance
- Cross-sell
- Upsell
- Discount efficiency
- Margin
- Incrementality
- Cannibalization
- A/B experiments

AI shall forecast

- Future campaign performance
- Revenue
- Redemption
- Conversion
- Margin
- Budget consumption
- Bundle demand
- Customer response

AI shall recommend

- Optimal discount
- Optimal promotion mechanic
- Optimal audience
- Optimal bundle
- Optimal channel
- Optimal timing
- Optimal campaign duration
- Budget reallocation
- Promotion termination
- Campaign expansion

AI Recommendation Confidence

Every recommendation should include a confidence indicator.

Example:

Recommendation Confidence: 87% — High

and an explanation of the principal drivers.

TICVAI should avoid presenting uncertain predictions as facts.

Where data quality or sample size is insufficient, the system should clearly indicate:

Insufficient evidence for reliable recommendation.

Reporting Data Architecture

Board 10 should not calculate analytics independently inside every channel.

I recommend a centralized architecture:

POS / B2C / B2B / Mobile / Kiosk / Partner

↓

Transaction + Promotion Events

↓

Central Analytics / Event Data Layer

↓

Promotion Attribution & KPI Engine

↓

AI Analytics Layer

↓

Board 10 / Power BI / Reports / API

This creates a single source of truth.

Metric Governance

Every major metric should have a governed definition.

For example:

Promotion-Influenced Revenue

Revenue from transactions where a promotion affected qualification, price, entitlement, or reward.

Discount Cost

Difference between the governed reference price and promotional selling price, adjusted according to configured financial rules.

Cross-Sell Revenue

Revenue from additional qualifying products purchased alongside or because of the primary product relationship.

Upsell Revenue

Incremental value generated when a customer moves from the reference product/price to a higher-value option.

Incremental Revenue

Estimated revenue above the modeled or measured no-promotion baseline.

Promotion ROI

Should use a centrally defined Finance-approved formula.

Softlabs should **not create different definitions of these metrics on different dashboards**.

Roles & Permissions

Support:

- Executive
- Marketing
- Commercial
- Revenue Management
- Finance
- Venue Management
- B2B
- CRM
- Data Analyst
- Auditor
- System Administrator

Sensitive data should be controlled.

For example:

Marketing may see:

Family Segment — Conversion 14.2%

without automatically receiving access to identifiable customer-level records.

Key Integrations

Boards 1-9

Promotion configuration and governance data.

Sales Channels

Transactions and offer interactions.

Ticketing Catalogue

Product dimensions.

CRM

Customer/segment dimensions.

Membership & Loyalty

Member behavior.

F&B / Retail

Cross-category performance.

Finance

Cost, margin, revenue and ROI.

Payment

Payment and bank-offer performance.

B2B / Reseller

Partner/channel attribution.

Power BI / Reporting

Enterprise reporting.

AI Layer

Prediction, attribution and optimization.

Important Architecture Principle

Board 10 closes the commercial feedback loop:

CONFIGURE

Boards 1–8

↓

GOVERN

Board 9

↓

SELL

TICVAI channels

↓

MEASURE

Board 10

↓

LEARN

AI Analytics

↓

RECOMMEND

Better audience / price / promotion / bundle

↓

SIMULATE & APPROVE

Board 9

↓

OPTIMIZE

Future campaign

This is the architecture I recommend for an **AI-first promotion platform**, rather than simply adding AI recommendation boxes to dashboards.

Board 10 Matrix Coverage

Board 10 consolidates the matrix requirements concerning:

- Promotion performance analytics
- Campaign financial performance
- Discount exposure
- Redemption performance
- Bundle performance

- Revenue contribution
- Margin impact
- ROI
- Cross-sell performance
- Upsell performance
- Customer/segment performance
- Channel performance
- Partner performance
- A/B test results
- AI campaign recommendations
- Historical behavior analysis
- Promotion optimization
- Reporting and management visibility

It also completes the analytical requirements introduced across the earlier promotion and bundle boards without recreating their configuration functionality.

Acceptance Criteria

Board 10 shall be considered complete when:

1. Promotion performance can be analyzed across all TICVAI channels.
2. Campaigns can be compared using consistent KPI definitions.
3. Gross, net, influenced, and estimated incremental revenue are distinguishable.
4. Discount exposure is measurable.
5. Promotion margin and profitability are measurable.
6. Redemption funnels can be analyzed.
7. Bundle performance can be analyzed.
8. Individual bundle component performance can be analyzed.
9. BOGO and reward performance can be measured.
10. Upsell revenue can be measured.
11. Cross-sell revenue can be measured.
12. Product attach rates can be analyzed.
13. Cross-category performance can be analyzed.
14. Customer segments can be compared.
15. Channels can be compared.
16. Partner performance can be analyzed.
17. Promotion ROI is calculated using a governed definition.
18. Incrementality can be estimated using appropriate available methodologies.
19. A/B/control-group data can be used for attribution where available.
20. Potential cannibalization can be identified.
21. AI can identify underperforming promotions.
22. AI can identify high-performing opportunities.
23. AI can recommend discount changes.

24. AI can recommend audience changes.
25. AI can recommend bundle and cross-sell opportunities.
26. Recommendations include expected commercial impact.
27. AI recommendations provide confidence/explanation.
28. Insufficient-data situations are clearly identified.
29. Users can move an AI recommendation into simulation/approval rather than directly modifying live campaigns.
30. Executives have a consolidated promotion intelligence view.
31. Natural-language analytics queries are supported.
32. Reports can be exported or integrated with enterprise BI subject to permission.
33. All major promotion KPIs use centrally governed definitions.
34. Data access respects tenant isolation and RBAC/PBAC.
35. The analytics architecture can consume new channels and promotion types without redesigning the board.

Board 10 is locked as: *Promotion Performance Analytics, Attribution & AI Optimization — 10 backend screens.*

With Board 10, the **10-board Promotions & Bundles Management module is functionally complete**: from campaign creation and rules through coupons, BOGO, bundles, dynamic availability, targeting, stacking, governance, and finally measurement/AI optimization.